Cross-intersecting systems beyond the classical threshold

Yang Huang* Yuejian Peng†

Abstract

The families $\mathcal{A}$ and $\mathcal{B}$ are cross-intersecting if $A \cap B \neq \emptyset$ for any $A \in \mathcal{A}$ and $B \in \mathcal{B}$. Let $t \geq 2$ and $k_1 \geq k_2 \geq \cdots \geq k_t$. We say that $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is an $(n, k_1, \ldots, k_t)$-cross-intersecting system if $\mathcal{F}_1 \subseteq \binom{[n]}{k_1}, \ldots, \mathcal{F}_t \subseteq \binom{[n]}{k_t}$ are non-empty pairwise cross-intersecting families. Let $M(n, k_1, \ldots, k_t)$ denote the maximum sum of sizes of families of an $(n, k_1, \ldots, k_t)$-cross-intersecting system. The case $t = 2$ was studied by Frankl and Tokushige (1992). Solving a problem of Shi, Frankl and Qian (2022), Huang, Peng and Wang (2024), as well as Zhang and Feng (2024) independently determined $M(n, k_1, \ldots, k_t)$ for all $n \geq k_1 + k_2$.

Observe that $n \geq k_1 + k_t$ is the most natural constraint. However, the previous methods in the literature are invalid in the range $n < k_1 + k_2$. In this paper, we overcome this obstacle and determine $M(n, k_1, \ldots, k_t)$ for $k_1 + k_3 \leq n < k_1 + k_2$. Furthermore, we characterize all extremal structures. This could be viewed as the first 'mixed-type' result about cross-intersection problem. Moreover, we introduce new concepts 'k-partner', 'parity' and 'corresponding k-set', and we develop some methods to determine whether two L-initial cross-intersecting families are maximal to each other. In addition, we prove that in an extremal L-initial $(n, k_1, \ldots, k_t)$-cross-intersecting system $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$, the sum $\sum_{i=1}^{t} |\mathcal{F}_i|$ can be expressed as a single variable function. We believe that our new result is of independent interest, and particularly, it plays an important role in the solution to the natural constraint $n \geq k_1 + k_t$.

Key words: Extremal combinatorics, Intersection theorems, cross-intersecting families
2010 Mathematics Subject Classification. 05D05, 05C65, 05D15.

1 Introduction

Extremal combinatorics studies the maximum or minimum size of a combinatorial object that has certain properties. One important class of extremal problems concerns intersection properties of set systems. We refer the reader to the surveys [11, 4]. This field is inspired by the fundamental result of Erdős, Ko and Rado [5]. We denote $[n] = \{1, 2, \ldots, n\}$. For $0 \leq k \leq n$, let $\binom{[n]}{k}$ be the family of all k-element subsets of $[n]$. A family $\mathcal{A}$ of subsets of $[n]$ is k-uniform if $\mathcal{A} \subseteq \binom{[n]}{k}$. A family $\mathcal{A}$ of sets is *intersecting* if $A \cap B \neq \emptyset$ for any $A, B \in \mathcal{A}$. Half a century ago, Erdős, Ko and Rado [5] showed that if $n > 2k$ and $\mathcal{A} \subseteq \binom{[n]}{k}$ is intersecting, then $|\mathcal{A}| \leq \binom{n-1}{k-1}$, with equality if and only if $\mathcal{A}$ consists of all k-subsets containing a fixed element. Various intersection theorems were proved in the literature, and some of them were applied to other fields, e.g., the theoretical computer science and discrete geometry. For instance, the famous Ahlswede–Khachatrian theorem [1] that determines the maximum size of a k-uniform family $\mathcal{A}$ where any two sets intersect in at least t elements plays a crucial role in the hardness-of-approximation theorem for the 'vertex cover' problem, proved by Dinur and Safra [3].

Two families $\mathcal{A}$ and $\mathcal{B}$ are called *cross-intersecting* if $A \cap B \neq \emptyset$ for any $A \in \mathcal{A}$ and $B \in \mathcal{B}$. In the case $\mathcal{A} = \mathcal{B}$, the problem for cross-intersecting families reduces to that for intersecting

*School of Mathematics, Hunan University; Moscow Institute of Physics and Technology; E-mail: 1060393815@qq.com.

†School of Mathematics, Hunan University; E-mail: ypeng1@hnu.edu.cn. The research is supported by National Natural Science Foundation of China (No. 12571363) and Natural Science Foundation of Hunan Province (Grant No. 2025JJ30003).

families. The families $\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_t$ are called *non-empty pairwise cross-intersecting* if every $\mathcal{F}_i$ is non-empty, and $\mathcal{F}_i$ and $\mathcal{F}_j$ are cross-intersecting for any $i \neq j$. In 1967, Hilton and Milner [17] determined the maximum sum of sizes of two cross-intersecting families.

Theorem 1.1 (Hilton–Milner [17]). *Let n and k be positive integers with $n \geq 2k$. If $\mathcal{A}, \mathcal{B} \subseteq \binom{[n]}{k}$ are non-empty cross-intersecting families, then*

$$|\mathcal{A}| + |\mathcal{B}| \leq \binom{n}{k} - \binom{n-k}{k} + 1.$$

Extending Theorem 1.1, Frankl and Tokushige [9] proved the following result.

Theorem 1.2 (Frankl–Tokushige [9]). *Let $\mathcal{A} \subseteq \binom{[n]}{k}$ and $\mathcal{B} \subseteq \binom{[n]}{\ell}$ be non-empty cross-intersecting families with $n \geq k + \ell$ and $k \geq \ell$. Then*

$$|\mathcal{A}| + |\mathcal{B}| \leq \binom{n}{k} - \binom{n-\ell}{k} + 1.$$

We refer the interested readers to [2, 6, 10, 12, 13, 26] for more related results for cross-intersecting families. In 2020, Shi, Frankl and Qian [24] proved a weighted extension of Theorem 1.2 by bounding the maximum of $|\mathcal{A}| + c|\mathcal{B}|$ for a constant $c > 0$. It is worth noting that an analogous weighted conclusion was independently obtained by Kupavskii [23]. Invoking the weighted version, Shi, Frankl and Qian [24] showed the following result for multi-families.

Theorem 1.3 (Shi–Frankl–Qian [24]). *Let n, k and t be positive integers with $n \geq 2k$ and $t \geq 2$. If $\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_t \subseteq \binom{[n]}{k}$ are non-empty pairwise cross-intersecting families, then*

$$\sum_{i=1}^{t} |\mathcal{F}_i| \leq \max\left\{\binom{n}{k} - \binom{n-k}{k} + t - 1, t\binom{n-1}{k-1}\right\}.$$

For given integers n, t and $k_1 \geq \cdots \geq k_t$, we say that $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is an $(n, k_1, \ldots, k_t)$-*cross-intersecting system* if $\mathcal{F}_1 \subseteq \binom{[n]}{k_1}, \ldots, \mathcal{F}_t \subseteq \binom{[n]}{k_t}$ are non-empty pairwise cross-intersecting families. Let $M(n, k_1, \ldots, k_t)$ be the maximum of $\sum_{i=1}^{t} |\mathcal{F}_i|$ over all $(n, k_1, \ldots, k_t)$-cross-intersecting systems. Moreover, a system $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is *extremal* if $\sum_{i=1}^{t} |\mathcal{F}_i| = M(n, k_1, \ldots, k_t)$.

A unified extension of Theorems 1.2 and 1.3 is to determine the value of $M(n, k_1, \ldots, k_t)$ for any integers $k_1 \geq \cdots \geq k_t$. In the range $n \geq k_1 + k_2$, the value of $M(n, k_1, \ldots, k_t)$ was determined by Huang, Peng and Wang [14], and independently by Zhang and Feng [27].

Theorem 1.4 ([14], [27]). *Let $\mathcal{F}_1 \subseteq \binom{[n]}{k_1}, \mathcal{F}_2 \subseteq \binom{[n]}{k_2}, \ldots, \mathcal{F}_t \subseteq \binom{[n]}{k_t}$ be non-empty pairwise cross-intersecting families with $t \geq 2$, $k_1 \geq k_2 \geq \cdots \geq k_t$. If $n \geq k_1 + k_2$, then*

$$\sum_{i=1}^{t} |\mathcal{F}_i| \leq \max\left\{\sum_{i=1}^{t}\binom{n-1}{k_i - 1}, \binom{n}{k_1} - \binom{n-k_t}{k_1} + \sum_{i=2}^{t}\binom{n-k_t}{k_i - k_t}\right\}.$$

Moreover, the extremal families are characterized as follows.

(i) For $n > k_1 + k_2$ or $n = k_1 + k_2 > k_1 + k_t$, the equality holds if and only if either there is a set $T \in \binom{[n]}{k_t}$ such that $\mathcal{F}_1 = \{F \in \binom{[n]}{k_1} \colon F \cap T \neq \emptyset\}$ and $\mathcal{F}_j = \{F \in \binom{[n]}{k_j} : T \subseteq F\}$ for every $j \in [2, t]$; or there exists $a \in [n]$ such that $\mathcal{F}_j = \left\{F \in \binom{[n]}{k_j} \colon a \in F\right\}$ for every $j \in [t]$.

(ii) For $t = 2$ and $n = k_1 + k_2$, the equality holds if and only if $\mathcal{F}_1 = \binom{[n]}{k_1} \setminus \{[n] \setminus A \colon A \in \mathcal{F}_2\}$.

(iii) For $t \geq 3$ and $n = k_1 + k_2 = k_1 + k_t$, if $k_1 > k_2$, then the equality holds if and only if $\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_t$ are full stars with the same center; if $k_1 = k_2$, then the equality holds if and only if $\mathcal{F}_1 = \mathcal{F}_2 = \cdots = \mathcal{F}_t$, and $\mathcal{F}_1$ is any intersecting family with size $\binom{n-1}{k_1-1}$.

This resolves an open problem of Shi, Frankl and Qian [24, Problem 4.3]. More generally, a weighted version for multi-families was recently studied by Huang and Peng [15].

Actually, the most natural constraint for this problem is $n \geq k_1 + k_t$, instead of $n \geq k_1 + k_2$. Indeed, if $n < k_1 + k_t$, then for any extremal $(n, k_1, \ldots, k_t)$-cross-intersecting system $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$, we must have $\mathcal{F}_1 = \binom{[n]}{k_1}$. In this case, the problem of determining $M(n, k_1, k_2, \ldots, k_t)$ reduces to that of determining $M(n, k_2, \ldots, k_t)$. However, when the constraint $n \geq k_1 + k_2$ is relaxed to $n \geq k_1 + k_t$, the methods previously established in [14, 27, 15, 24] cease to be applicable.

For cross-intersecting families $\mathcal{A} \subseteq \binom{[n]}{k}$ and $\mathcal{B} \subseteq \binom{[n]}{\ell}$, we say that $\mathcal{A}$ and $\mathcal{B}$ are *freely cross-intersecting* whenever $n < k + \ell$; otherwise, we say that $\mathcal{A}$ and $\mathcal{B}$ are *non-freely cross-intersecting*. In addition, we say that the cross-intersecting system $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is of *mixed type* if it contains at least two families that are freely cross-intersecting, and also contains at least two families that are non-freely cross-intersecting; otherwise, we say that $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is of *non-mixed type*.

The aforementioned references considered the intersection problem only for the non-mixed type systems, which require the condition $n \geq k_1 + k_2$. It would be interesting to study this problem for mixed type systems admitting the condition $n < k_1 + k_2$. In this paper, we give the first mixed type result in the range $k_1 + k_3 \leq n < k_1 + k_2$. Given families $\mathcal{F}_1 \subseteq \binom{[n]}{k_1}, \ldots, \mathcal{F}_t \subseteq \binom{[n]}{k_t}$ with $k_1 \geq k_2 \geq \cdots \geq k_t$, if $k_1 + k_3 \leq n < k_1 + k_2$, then only $\mathcal{F}_1$ and $\mathcal{F}_2$ are freely cross-intersecting in the mixed type system $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$. We begin with the following examples.

Example 1.5. *Let $k_1 \geq k_2 \geq \cdots \geq k_t$ and $k_1 + k_3 \leq n < k_1 + k_2$. For each $i \in [t]$, let*

$$\mathcal{G}_i = \left\{ G \in \binom{[n]}{k_i} : 1 \in G \right\}.$$

Example 1.6. *Let $k_1 \geq k_2 \geq \cdots \geq k_t$ and $k_1 + k_3 \leq n < k_1 + k_2$. For $i \in \{1, 2\}$, let*

$$\mathcal{H}_i = \left\{ H \in \binom{[n]}{k_i} : H \cap [k_t] \neq \emptyset \right\},$$

and for every $i \in [3, t]$, let

$$\mathcal{H}_i = \left\{ H \in \binom{[n]}{k_i} : [k_t] \subseteq H \right\}.$$

In this paper, we shall show that under the constraint $k_1 + k_3 \leq n < k_1 + k_2$, an $(n, k_1, \ldots, k_t)$-cross-intersecting system $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal if and only if it is isomorphic to the systems in Examples 1.5 or 1.6, with one more exception possibly. For a family $\mathcal{A}$, we denote $\overline{\mathcal{A}} := \{[n] \setminus A : A \in \mathcal{A}\}$.

Theorem 1.7 (Main result)**.** *Let $t \geq 3$, $k_1 \geq k_2 \geq \cdots \geq k_t$ and $k_1 + k_3 \leq n < k_1 + k_2$. If $\mathcal{F}_1 \subseteq \binom{[n]}{k_1}, \mathcal{F}_2 \subseteq \binom{[n]}{k_2}, \ldots, \mathcal{F}_t \subseteq \binom{[n]}{k_t}$ are non-empty pairwise cross-intersecting families, then*

$$\sum_{i=1}^{t} |\mathcal{F}_i| \leq \max \left\{ \sum_{i=1}^{t} \binom{n-1}{k_i - 1}, \sum_{i=1}^{2} \left(\binom{n}{k_i} - \binom{n-k_t}{k_i} \right) + \sum_{i=3}^{t} \binom{n-k_t}{k_i - k_t} \right\}.$$

If $t = 4$, $k_1 = k_2$, $k_3 = k_4$ and $n = k_1 + k_3$, then the equality holds if and only if $\mathcal{F}_3 = \mathcal{F}_4$ is an intersecting family, and $\mathcal{F}_1 = \mathcal{F}_2 = \binom{[n]}{k_1} \setminus \overline{\mathcal{F}_3}$. Otherwise, the equality holds if and only if $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is isomorphic to $(\mathcal{G}_1, \ldots, \mathcal{G}_t)$ in Example 1.5 or $(\mathcal{H}_1, \ldots, \mathcal{H}_t)$ in Example 1.6.

It is worth mentioning that the contribution of this paper not only lies in studying the case where $n < k_1 + k_2$, but also provides a feasible approach toward ultimately resolving the most natural constraint $n \geq k_1 + k_t$. The key ingredient in our method is based on utilizing the results involving the 'k-partner', 'parity' and 'corresponding k-set', which are of independent interest.

1.1 Sketch of the proof

To determine $M(n, k_1, \ldots, k_t)$, a result of Kruskal and Katona (Theorem 2.1) allows us to consider only families $\mathcal{F}_i$ whose members are the first $|\mathcal{F}_i|$ in lexicographic order (we call them L-initial families) in an extremal $(n, k_1, \ldots, k_t)$-cross-intersecting system $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$. For an L-initial family $\mathcal{F}$, we call the last member (under lexicographic order) of $\mathcal{F}$ the *ID* of $\mathcal{F}$. In [15], we came up with a new method different from [14, 27] and bounded $\sum_{i=1}^{t} c_i|\mathcal{F}_i|$ (where c_i, $1 \le i \le t$ are positive constants) by a function $f_i(R)$, where R is the ID of $\mathcal{F}_i$, and showed that $-f_i(R)$ has 'unimodality'. The method and results in our previous work [15] provide some foundation for this paper, and we need new ingredients to overcome the difficulty raised by weakening the condition on n. Since the condition on n is relaxed to $k_1+k_3 \le n < k_1+k_2$, $\mathcal{F}_1$ and $\mathcal{F}_2$ are freely cross-intersecting. When we try to bound $\sum_{i=1}^{t} |\mathcal{F}_i|$ by a function, there are two free variables I_1 (the ID of $\mathcal{F}_1$) and I_2 (the ID of $\mathcal{F}_2$). This causes more difficulty to analyze properties of the corresponding function, comparing to the problem in [15]. To overcome this difficulty, we introduce new concepts 'k-partner' , 'parity' and 'corresponding k-set', develop some rules to determine whether a pair of L-initial cross-intersecting families are maximal to each other, and prove one crucial property that if $(\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_t)$ is an extremal L-initial with IDs I_1, I_2 of $\mathcal{F}_1, \mathcal{F}_2$, respectively, then I_2 is the corresponding k_2-set of I_1. This discovery allows us to bound $\sum_{i=1}^{t} |\mathcal{F}_i|$ by a single variable function $f(I_1)$. Another crucial and challenging part is to bound $f(I_1)$. Comparing to the function in [15], we need to overcome more difficulties in dealing with function $f(I_1)$ since there are more 'mysterious' terms to be taken care of. We take advantage of some properties of function $f_i(R)$ obtained in [15] and come up with some new strategies in estimating the change $f(I_1') - f(I_1)$ as the ID of $\mathcal{F}_1$ increases from I_1 to I_1'. The new concepts introduced in this paper—including the 'k-partner', 'parity', and 'corresponding k-set'—along with the developed rules for determining whether a pair of L-initial cross-intersecting families are maximal with respect to each other, will serve as a crucial foundation for the solution to the most general constraint $n \ge k_1 + k_t$ in the follow-up paper [16]. Key results, such as Lemma 4.17, are also essential to this framework.

1.2 Organization of the paper

In Section 2, we introduce the new concepts 'k-partner' and 'parity', and give some results to determine whether a pair of L-initial cross-intersecting families are maximal to each other. These results form a crucial foundation for addressing the general case where $n \ge k_1 + k_t$. Section 3 summarizes key results from our earlier work on the non-mixed type, which will be utilized in the present paper. In Section 4, we present the proof of Theorem 1.7, assuming the validity of Lemma 4.17. Finally, in the last section, we provide the proof of Lemma 4.17, which will also play an essential role in subsequent work [16].

2 Preliminaries

In this section, we introduce new concepts 'k-partner' and 'parity', develop some rules to determine whether a pair of L-initial cross-intersecting families are maximal to each other.

When we write a set $A = \{a_1, a_2, \ldots, a_s\} \subseteq [n]$, we always assume that $a_1 < a_2 < \ldots < a_s$. We define the interval $[a, b]$ as follows: If $a < b$, then $[a, b] = \{a, a+1, \ldots, b\}$. If $a = b$, then $[a, b] = \{a\} = \{b\}$. If $a > b$, then $[a, b] = \emptyset$. For $A \subseteq [n]$, let $\max A$ and $\min A$ denote the largest element and the smallest element of A, respectively. Let A and B be finite subsets of the set of positive integers. We say that A is no more than B in *lexicographic (lex for short) order*, denoted by $A \prec B$, if either $A \supseteq B$ or $\min(A \setminus B) < \min(B \setminus A)$. In particular, $A \prec A$. For two subsets $A, B \subset [n]$, we write $A \precneqq B$ if $A \prec B$ and $A \ne B$. Let $\mathcal{L}([n], r, k)$ denote the first r

subsets in $\binom{[n]}{k}$ in the lex order. Given a set R, denote

$$\mathcal{L}([n], R, k) = \Big\{F \in \binom{[n]}{k} : F \prec R\Big\}. \tag{1}$$

Whenever the underlying set $[n]$ is clear, we shall ignore it and write $\mathcal{L}(R,k)$, $\mathcal{L}(r,k)$ for short. Let $\mathcal{F} \subseteq \binom{[n]}{k}$ be a family, we say $\mathcal{F}$ is *L-initial* if $\mathcal{F} = \mathcal{L}([n], |\mathcal{F}|, k)$. We call the last member of $\mathcal{F}$ the *ID* of $\mathcal{F}$.

The well-known Kruskal–Katona theorem [20, 22] will allow us to consider only L-initial families. (If we only consider the size of an $(n, k_1, \ldots, k_t)$-cross-intersecting system.) An equivalent formulation of this result was given in [8, 18] as follows.

Theorem 2.1 (Kruskal–Katona theorem [8, 18])**.** *For $\mathcal{A} \subseteq \binom{[n]}{k}$ and $\mathcal{B} \subseteq \binom{[n]}{l}$, if $\mathcal{A}$ and $\mathcal{B}$ are cross-intersecting, then $\mathcal{L}(|\mathcal{A}|, k)$ and $\mathcal{L}(|\mathcal{B}|, l)$ are cross-intersecting as well.*

Let F and H be two subsets of $[n]$ with size $|F| = f$ and $|H| = h$. We say that F and H *strongly intersect* at their last element if there is an element q such that $F \cap H = \{q\}$ and $F \cup H = [q]$. We also say F is the *partner* of H (or H is the *partner* of F). In [15], the authors proved the following proposition.

Proposition 2.2 (Huang-Peng [15])**.** *Let a, b, n be positive integers and $a + b \le n$. For $P \subseteq [n]$ with $|P| \le a$, let Q be the partner of P. Then $\mathcal{L}(Q, b)$ is the maximum b-uniform family that is cross-intersecting with $\mathcal{L}(P, a)$.*

In [15], we worked on non-mixed type: Let $t \ge 2$, $k_1 \ge k_2 \ge \cdots \ge k_t$ and $n \ge k_1 + k_2$ and families $\mathcal{A}_1 \subseteq \binom{[n]}{k_1}, \mathcal{A}_2 \subseteq \binom{[n]}{k_2}, \ldots, \mathcal{A}_t \subseteq \binom{[n]}{k_t}$ be non-empty pairwise cross-intersecting. Let R be the ID of $\mathcal{A}_1$, and T be the partner of R. In view of [15], one important ingredient is that by Proposition 2.2, $\sum_{i=1}^{t} |\mathcal{A}_i|$ can be bounded by a function of R as follows.

$$f(R) = \sum_{j=1}^{t} |\mathcal{A}_j| \le |\mathcal{L}(R, k_1)| + \sum_{j=2}^{t} |\mathcal{L}(T, k_j)|.$$

Let $\mathcal{F}_1, \ldots, \mathcal{F}_t$ be the families described in Theorem 1.7. By Theorem 2.1, to prove the quantitative part of Theorem 1.7 we may also assume that $\mathcal{F}_i$ is L-initial, that is, for each $i \in [t]$, we have $\mathcal{F}_i = \mathcal{L}(I_i, k_i)$, where I_i is the ID of $\mathcal{F}_i$. However, the condition on n is relaxed to $k_1 + k_3 \le n < k_1 + k_2$, so $\mathcal{F}_1 \subseteq \binom{[n]}{k_1}$ and $\mathcal{F}_2 \subseteq \binom{[n]}{k_2}$ are freely cross-intersecting. When we try to bound $\sum_{i=1}^{t} |\mathcal{F}_i|$ by a function, there are two free variables I_1 (the ID of $\mathcal{F}_1$) and I_2 (the ID of $\mathcal{F}_2$). This causes more difficulty to analyze properties of the corresponding function, comparing to the problem in [15]. To overcome this difficulty, some new ideas are needed.

2.1 Partner and Parity

Now, we introduce new concepts 'k-partner' and 'parity', and develop some criteria to determine whether a pair of L-initial cross-intersecting families are maximal to each other.

To start with, we need to define the following concepts.

- $\mathcal{F}$ **is maximal to** $\mathcal{G}$**.** Let $\mathcal{F} \subseteq \binom{[n]}{f}$ and $\mathcal{G} \subseteq \binom{[n]}{g}$ be cross-intersecting. We say that $\mathcal{F}$ *is maximal to* $\mathcal{G}$ if $\mathcal{F}' \subseteq \binom{[n]}{f}$ and $\mathcal{G}$ are cross-intersecting with $\mathcal{F} \subseteq \mathcal{F}'$, then $\mathcal{F}' = \mathcal{F}$.
- $(\mathcal{F}, \mathcal{G})$ **is maximal.** The pair $(\mathcal{F}, \mathcal{G})$ is maximal if and only if $\mathcal{F}$ and $\mathcal{G}$ are maximal with respect to each other.
- F **is maximal to** G**.** Let F and G be two subsets of $[n]$. We say that F *is maximal to* G if there are two L-initial families $\mathcal{F} \subseteq \binom{[n]}{|F|}$ and $\mathcal{G} \subseteq \binom{[n]}{|G|}$ with IDs F and G respectively such that $\mathcal{F}$ is maximal to $\mathcal{G}$.

- (F, G) **is maximal.** The pair (F, G) is maximal if and only if F and G are maximal with respect to each other.

- **maximal pair families.** We say two families $\mathcal{A}_1$ and $\mathcal{A}_2$ are *maximal pair families* if $|\mathcal{A}_1| = |\mathcal{A}_2|$ and for every $A_1 \in \mathcal{A}_1$, there is a unique $A_2 \in \mathcal{A}_2$ such that (A_1, A_2) is maximal.

- Let $F = \{x_1, x_2, \ldots, x_k\} \subseteq [n]$. We denote

$$\ell(F) = \begin{cases} \max\{x : [n-x+1, n] \subseteq F\}, & \text{if } \max F = n; \\ 0, & \text{if } \max F < n. \end{cases}$$
$$F^{\mathrm{t}} = [n - \ell(F) + 1, n].$$

- $F < G$**.** For two sets $F, G \subseteq [n]$, we write $F < G$ if $F \precneqq G$ and there is no $F' \subseteq [n]$ such that $|F'| = |F|$ and $F \precneqq F' \precneqq G$.

- k**-partner.** Let F and H be two subsets of $[n]$ and partners to each other with $|F| = f$ and $|H| = h$. Let $k \leq n - f$ be an integer, we define the *k-partner* K of F as follows. For $k = h$, let $K = H$. If $k \neq h$, then let $K \subseteq [n]$ be the k-set such that $K < H$.

- **parity.** Let h_1 and h_2 be positive integers with $h_1 \leq h_2$, H_1 and H_2 be subsets of $[n]$ with sizes h_1 and h_2, respectively. We say H_1 is the *h_1-parity* of H_2 (or H_2 is the h_2-parity of H_1) if $H_1 \setminus H_1^{\mathrm{t}} = H_2 \setminus H_2^{\mathrm{t}}$ and $\ell(H_2) - \ell(H_1) = h_2 - h_1$.

Note that if $k > h$, then $K = H \cup \{n-k+h+1, \ldots, n\}$. Indeed, since F and H intersect at their last element, $n' = \max H = f + h - 1 < n - k + h$. This implies $K^{\mathrm{t}} = \{n-k+h+1, \ldots, n\}$ and $|K| = |H \cup \{n-k+h+1, \ldots, n\}| = k$. From the definition of k-partner, we have the following fact and remark.

Fact 2.3. *Let a, b, c, n be positive integers, $a \geq b$ and $n \geq a + c$, and let C be a c-subset of $[n]$. Suppose that A is the a-partner of C and B is the b-partner of C, then $B \prec A$ or A is the a-parity of B.*

Remark 2.4. *Let $F \subseteq [n]$ with $|F| = f$ and $k \leq n - f$. Suppose that H is the partner of F, and K is the k-partner of F, then we have $\mathcal{L}(H, k) = \mathcal{L}(K, k)$.*

By Proposition 2.2 and Remark 2.4, we have the following fact.

Fact 2.5. *Let a, b, n be positive integers and $n \geq a + b$. For $A \subseteq [n]$ with $|A| = a$, let B be the b-partner of A, then $\mathcal{L}(B, b)$ and $\mathcal{L}(A, a)$ are cross-intersecting, moreover, $\mathcal{L}(B, b)$ is maximal to $\mathcal{L}(A, a)$.*

Note that families $\mathcal{L}(A, a)$ and $\mathcal{L}(B, b)$ which are mentioned in Fact 2.5 may not be maximal cross-intersecting, since we don't know whether $\mathcal{L}(A, a)$ is maximal to $\mathcal{L}(B, b)$. For example, let $n = 9$, $a = 3$, $b = 4$ and $A = \{2, 4, 7\}$. Then the b-partner of A is $\{1, 3, 4, 9\}$. Although $\mathcal{L}(\{1, 3, 4, 9\}, 4)$ is maximal to $\mathcal{L}(\{2, 4, 7\}, 3)$, $\mathcal{L}(\{1, 3, 4, 9\}, 4)$ and $\mathcal{L}(\{2, 4, 7\}, 3)$ are not maximal cross-intersecting families since $\mathcal{L}(\{2, 4, 7\}, 3) \subsetneq \mathcal{L}(\{2, 4, 9\}, 3)$, and $\mathcal{L}(\{2, 4, 9\}, 3)$ and $\mathcal{L}(\{1, 3, 4, 9\}, 4)$ are cross-intersecting families.

Fact 2.6. *Let a, b, n be positive integers and $n \geq a + b$. Let $\mathcal{L}(A, a)$ and $\mathcal{L}(B, b)$ be cross-intersecting families. If $\{1, n - a + 2, \ldots, n\} \prec A$, then $B \prec \{1, n - b + 2, \ldots, n\}$.*

Fact 2.7. *Let $F \subseteq [n]$ with $|F| = f$ and $k \leq n - f$. Suppose that K and K' are the k-partners of F and $F \setminus F^{\mathrm{t}}$, respectively, then $K = K'$.*

Proof. If $\ell(F) = 0$, then we are done. Suppose $\ell(F) > 0$. Write $F = \{x_1, \ldots, x_y\} \cup F^{\mathrm{t}}$ and $C := H \cap [x_y - 1]$. Let H and H' be the partners of F and $F \setminus F^{\mathrm{t}}$, respectively. Then

$$H = C \cup [x_y + 1, n - \ell(F)] \cup \{n\}, \quad H' = C \cup \{x_y\}.$$

Thus H and H' have the same initial part C, and the definition of the k-partner only requires us to distinguish the three cases $k < |H'|$,$k = |H'|$, and $k > |H'|$.

Since $|H| = n - f + 1 > n - f$, $k < |H|$. If $k \leq |H'| - 1$, then $K = K'$, as desired; if $k = |H'|$, then $K' = H'$ and $K = C \cup \{x_y\} = H'$, as desired; if $k > |H'|$, then $K' = H' \cup [n - k + |H'| + 1, n]$ and $K = C \cup \{x_y\} \cup [n - k + |H'| + 1, n] = K'$, as desired. □

Frankl and Kupavskii [7] gave a sufficient condition for a pair of maximal cross-intersecting families, and a necessary condition for a pair of maximal cross-intersecting families as stated below.

Proposition 2.8 (Frankl–Kupavskii [7])**.** *Let a, b, n be positive integers with $n \geq a + b$. Let P and Q be non-empty subsets of $[n]$ with $|P| \leq a$ and $|Q| \leq b$. If Q is the partner of P, then $\mathcal{L}(P, a)$ and $\mathcal{L}(Q, b)$ are maximal cross-intersecting families. Conversely, if $\mathcal{L}(A, a)$ and $\mathcal{L}(B, b)$ are maximal cross-intersecting families with $|A| = a$ and $|B| = b$, let j be the smallest element of $A \cap B$, $P = A \cap [j]$ and $Q = B \cap [j]$. Then $\mathcal{L}(P, a) = \mathcal{L}(A, a)$, $\mathcal{L}(Q, b) = \mathcal{L}(B, b)$ and P, Q satisfy the following conditions: $|P| \leq a$, $|Q| \leq b$, and Q is the partner of P.*

Based on Proposition 2.8, we point out a necessary and sufficient condition for a pair of maximal cross-intersecting families in terms of their IDs.

Lemma 2.9. *Let A and B be non-empty subsets of $[n]$ with $|A| + |B| \leq n$. Let $A' = A \setminus A^{\mathrm{t}}$ and $B' = B \setminus B^{\mathrm{t}}$. Then (A, B) is maximal if and only if A' is the partner of B'.*

Proof. Let $|A| = a$ and $|B| = b$. Since $A' = A \setminus A^{\mathrm{t}}$ and $B' = B \setminus B^{\mathrm{t}}$, $\mathcal{L}(A, a) = \mathcal{L}(A', a)$ and $\mathcal{L}(B, b) = \mathcal{L}(B', b)$. First we show the sufficiency. Suppose that A' is the partner of B'. Since $|A'| \leq |A|$ and $|B'| \leq |B|$, by Proposition 2.8, $\mathcal{L}(A', a)$ and $\mathcal{L}(B', b)$ are maximal cross-intersecting families. Thus $\mathcal{L}(A, a)$ and $\mathcal{L}(B, b)$ are maximal cross-intersecting families, in other words, (A, B) is maximal. Next, we show the necessity. Suppose that (A, B) is maximal. Let j be the smallest element of $A \cap B$, $P = A \cap [j]$ and $Q = B \cap [j]$. By Proposition 2.8, $\mathcal{L}(A, a) = \mathcal{L}(P, a)$, $\mathcal{L}(B, b) = \mathcal{L}(Q, b)$; $|P| \leq a$, $|Q| \leq b$ and P is the partner of Q. Since $\mathcal{L}(A, a) = \mathcal{L}(P, a)$ and $P \subseteq A$, $A = P \cup \{n - a + |P| + 1, \ldots, n\}$. Similarly, $B = Q \cup \{n - b + |Q| + 1, \ldots, n\}$. Note that $|P| \leq a$, $|Q| \leq b$ and $n \geq a + b$. Then $|Q| \leq n - a$ and $|P| \leq n - b$. Since P is the partner of Q and $\max P = \max Q = j$, $|P| + |Q| = j + 1$. So $j + 1 - |P| = |Q| \leq n - a$ and $j + 1 - |Q| = |P| \leq n - b$. Consequently, $j \leq n - a + |P| - 1$ and $j \leq n - b + |Q| - 1$. Hence, $P = A \setminus A^{\mathrm{t}} = A'$ and $Q = B \setminus B^{\mathrm{t}} = B'$. Since P is the partner of Q, A' is the partner of B'. □

Fact 2.10. *Let a, b, n be positive integers and $n \geq a + b$. Suppose that A is an a-subset of $[n]$, and B is the b-partner of A. Let A' be the a-partner of B, then (A', B) is maximal. Moreover, $A \prec A'$.*

Proof. By Fact 2.5, B is maximal to A. So $\mathcal{L}([n], A, a)$ and $\mathcal{L}([n], B, b)$ are cross-intersecting. Since A' is the a-partner of B, by Fact 2.5 again, $\mathcal{L}([n], A', a)$ and $\mathcal{L}([n], B, b)$ are cross-intersecting and A' is maximal to B. So $\mathcal{L}([n], A, a) \subseteq \mathcal{L}([n], A', a)$, i.e., $A \prec A'$. □

Fact 2.11. *Let a, b, k, n be positive integers and $n \geq \max\{a + k, b + k\}$. Let A and B be two subsets of $[n]$ with sizes $|A| = a$ and $|B| = b$. Suppose that K_a and K_b are the k-partners of A and B respectively. If $A \prec B$, then $K_b \prec K_a$. In particular, if A is the a-parity of B, then $K_a = K_b$.*

Proof. Let K_A and K_B be the partners of A and B, respectively. Since $A \prec B$, $K_B \prec K_A$. Therefore, by the definition of k-partner directly, $K_b \prec K_a$, as required. For the second claim, since A is the a-parity of B, $A \setminus A^{\mathrm{t}} = B \setminus B^{\mathrm{t}}$. Let F be the k-partner of $A \setminus A^{\mathrm{t}}$. So F is the k-partner of $B \setminus B^{\mathrm{t}}$ as well. By Fact 2.7, $K_a = F$ and $K_b = F$. So $K_a = K_b$, as required. □

Definition 2.12. *Let $h_1 \leq h_2$. For two families $\mathcal{H}_1 \subseteq \binom{[n]}{h_1}$ and $\mathcal{H}_2 \subseteq \binom{[n]}{h_2}$, we say that $\mathcal{H}_1$ is the h_1-parity of $\mathcal{H}_2$ (or $\mathcal{H}_2$ is the h_2-parity of $\mathcal{H}_1$) if the following properties hold:*

(i) for any $H_1 \in \mathcal{H}_1$, the h_2-parity of H_1 exists and must be in $\mathcal{H}_2$;

(ii) for any $H_2 \in \mathcal{H}_2$, either H_2 has no h_1-parity or its h_1-parity is in $\mathcal{H}_1$.

From the definition of parity, if a set has a k-parity, then it has the unique one.

Proposition 2.13. *Let f, g, h, n be positive integers with $f \geq g$ and $n \geq f + h$. Let*

$$\mathcal{F} = \left\{ F \in \binom{[n]}{f} : \text{ there exists } H \in \binom{[n]}{h} \text{ such that } (F, H) \text{ is maximal} \right\},$$
$$\mathcal{G} = \left\{ G \in \binom{[n]}{g} : \text{ there exists } H \in \binom{[n]}{h} \text{ such that } (G, H) \text{ is maximal} \right\}.$$

Let $\mathcal{H}_{\mathcal{F}} \subseteq \binom{[n]}{h}$ and $\mathcal{H}_{\mathcal{G}} \subseteq \binom{[n]}{h}$ be the families such that $\mathcal{F}$ and $\mathcal{H}_{\mathcal{F}}$ are maximal pair families and $\mathcal{G}$ and $\mathcal{H}_{\mathcal{G}}$ are maximal pair families. Then the following properties hold:

(i) $\mathcal{F}$ is the f-parity of $\mathcal{G}$;

(ii) $\mathcal{H}_{\mathcal{G}} \subseteq \mathcal{H}_{\mathcal{F}}$;

(iii) for any $G \in \mathcal{G}$, let $F \in \mathcal{F}$ be the f-parity of G and let $H \in \mathcal{H}_{\mathcal{F}}$ be such that (F, H) is maximal, then (G, H) is maximal.

Proof. If $f = g$, then $\mathcal{F} = \mathcal{G}$ and $\mathcal{H}_{\mathcal{G}} = \mathcal{H}_{\mathcal{F}}$, we are done. We next consider the case that $f = g + s$ for some $s \geq 1$.

First, we are going to prove (i). Let $G \in \mathcal{G}$. Then there is the unique $H \in \mathcal{H}_{\mathcal{G}}$ such that (G, H) is maximal. Let $G' = G \setminus G^{\mathrm{t}} = G \setminus \{n - \ell(G) + 1, \ldots, n\}$ and $H' = H \setminus H^{\mathrm{t}} = H \setminus \{n - \ell(H) + 1, \ldots, n\}$. Then, by Lemma 2.9, G' and H' are partners of each other. So $\max G' = |G'| + |H'| - 1 = g - \ell(G) + |H'| - 1 = f - s - \ell(G) + |H'| - 1 \leq f + h - s - \ell(G) - 1 \leq n - \ell(G) - s - 1$. Let $F = G' \cup \{n - \ell(G) + 1 - s, \ldots, n\}$. Then $\max G' \leq n - \ell(G) - s - 1$ implies $F^{\mathrm{t}} = \{n - \ell(G) + 1 - s, \ldots, n\}$. Then $G \subseteq F$, $|F| = f$ and $\ell(F) - \ell(G) = f - g = s$. By the definition of parity, F is the f-parity of G. Let $F' = F \setminus F^{\mathrm{t}}$. So $F' = G'$. Furthermore, F' and H' are partners of each other. By Lemma 2.9 again, (F, H) is maximal. So $F \in \mathcal{F}$. Then to prove (i), we owe to confirm Definition 2.12 (ii). Let $F \in \mathcal{F}$. If F dose not have g-parity, then we are done. Assume that G is the g-parity of F. Then $G \setminus G^{\mathrm{t}} = F \setminus F^{\mathrm{t}}$. Since $F \in \mathcal{F}$, there exists H in $\mathcal{H}_{\mathcal{F}}$ such that (F, H) is maximal. By Lemma 2.9, $F \setminus F^{\mathrm{t}}$ is the partner of $H \setminus H^{\mathrm{t}}$. So $G \setminus G^{\mathrm{t}}$ is the partner of $H \setminus H^{\mathrm{t}}$ as well. By Lemma 2.9 again, (G, H) is maximal. So $G \in \mathcal{G}$. This implies that $\mathcal{F}$ is the f-parity of $\mathcal{G}$, as desired.

Next, we are going to prove (ii). Let $H \in \mathcal{H}_{\mathcal{G}}$. Then there exists $G \in \mathcal{G}$ such that (G, H) is maximal. By (i), there exists $F \in \mathcal{F}$ such that F is the f-parity of G. By Lemma 2.9, $G \setminus G^{\mathrm{t}}$ is the partner of $H \setminus H^{\mathrm{t}}$. Since F is the f-parity of G, $G \setminus G^{\mathrm{t}} = F \setminus F^{\mathrm{t}}$. So $F \setminus F^{\mathrm{t}}$ is the partner of $H \setminus H^{\mathrm{t}}$. By Lemma 2.9 again, (F, H) is maximal as well. This implies $\mathcal{H}_{\mathcal{G}} \subseteq \mathcal{H}_{\mathcal{F}}$, as required.

At last, we are going to prove (iii). Let $G \in \mathcal{G}$, and let F be the f-parity of G. By (i), $F \in \mathcal{F}$. So there exists $H \in \mathcal{H}_{\mathcal{F}}$ such that (F, H) is maximal. By Lemma 2.9, $F \setminus F^{\mathrm{t}}$ is the partner of $H \setminus H^{\mathrm{t}}$. Since F is the f-parity of G, $G \setminus G^{\mathrm{t}} = F \setminus F^{\mathrm{t}}$. So $G \setminus G^{\mathrm{t}}$ is the partner of $H \setminus H^{\mathrm{t}}$. By Lemma 2.9 again, (G, H) is maximal, as desired. □

Proposition 2.14. *Let $n = k + \ell$, A be any k-subset of $[n]$. Then there is an ℓ-set $B \subset [n]$ such that (A, B) is maximal. Moreover, for any $A \in \binom{[n]}{k}$, let B be the ℓ-set such that (A, B) is maximal, then we have $M(n, k, \ell) = |\mathcal{L}(A, k)| + |\mathcal{L}(B, \ell)| = \binom{n}{\ell}$.*

Proof. Let $A \in \binom{[n]}{k}$ and let B' be the partner of $A \setminus A^{\mathrm{t}}$. Since $n = k + \ell$, $|B'| \le \ell$ and $\max B' < n$, in other words, $\ell(B') = 0$. If $|B'| = \ell$, then let $B = B'$. If $|B'| < \ell$, then let $B = B' \cup [n-\ell+|B'|+1, n]$. By Lemma 2.9, (A, B) is maximal. This implies $\mathcal{L}(B, \ell) = \binom{[n]}{\ell} \setminus \overline{\mathcal{L}(A, k)}$ since $n = k + \ell$. Thus, $M(n, k, \ell) = |\mathcal{L}(A, k)| + |\mathcal{L}(B, \ell)| = \binom{n}{\ell}$, we are done. □

3 Results of non-mixed type

The purpose of this section is to recall the part of the one-variable method from [15] that will be used later. Once the ID R of the first family is fixed, the largest possible choices of the remaining non-freely cross-intersecting families are determined by the appropriate partners of R. Thus the total size becomes a function of the single variable R. The results below first describe the local variation of this function and then identify its possible extremal values. Denote

$$\mathbb{R}_1 = \Big\{ R \in \binom{[n]}{k_1} : \{1, n-k_1+2, \dots, n\} \prec R \prec \{k_t, n-k_1+2, \dots, n\} \Big\}.$$

Two related functions are defined in [15] as follows.

Definition 3.1. *Let $t \ge 2$, $k_1, k_2, \dots, k_t$ be positive integers with $k_1 \ge k_2 \ge \cdots \ge k_t$ and $n \ge k_1 + k_2$. Let R and R' belong to $\mathbb{R}_1$ such that $R \prec R'$. For each $j \in [2, t]$, let T_j and T'_j be the k_j-partners of R and R', respectively. We define*

$$\alpha(R, R') = |\mathcal{L}(R', k_1)| - |\mathcal{L}(R, k_1)|, \tag{2}$$

$$\beta(R, R') = \sum_{j=2}^{t} \big(|\mathcal{L}(T_j, k_j)| - |\mathcal{L}(T'_j, k_j)| \big). \tag{3}$$

The quantities α and β have the following gain–loss interpretation. Define

$$\Phi(R) := |\mathcal{L}(R, k_1)| + \sum_{j=2}^{t} |\mathcal{L}(T_j(R), k_j)|,$$

where $T_j(R)$ denotes the k_j-partner of R. When the ID of the first family moves from R to R', the first family gains $\alpha(R, R')$ members, whereas the partner families lose altogether $\beta(R, R')$ members. Consequently,

$$\Phi(R') - \Phi(R) = \alpha(R, R') - \beta(R, R').$$

Proposition 3.2 computes this difference for two consecutive IDs. Proposition 3.3 and Lemma 3.5 then allow the same local computation to be transferred between suitable lexicographic intervals.

Proposition 3.2 (Huang-Peng [15])**.** *Let $n \ge k_1 + k_2$ and $k_1 \ge k_2 \ge \cdots \ge k_t$. Let $F, G \in \mathbb{R}_1$ with $F < G$ and $\max G = q$. Then*
(i) $\alpha(F, G) = 1$.
(ii) $\beta(F, G) = \sum_{j=2}^{t} \binom{n-q}{k_j-(q-k_1)}$.
(iii) If $n = k_1 + k_j$ holds for any $j \in [t] \setminus \{1\}$, then $\beta(F, G) = t - 1$; otherwise, $\beta(F, G)$ decreases when q strictly increases until $\beta(F, G) = 0$.

In particular, when $F < G$, the first family gains exactly one member. Hence the sign of $\Phi(G) - \Phi(F)$ is governed by $1 - \beta(F, G)$. Proposition 3.2(iii) describes how this quantity changes as $\max G$ increases. This is the discrete unimodality phenomenon used in our earlier work.

Proposition 3.3 (Huang-Peng [15])**.** *Let $n \ge k_1 + k_2$ and $k_1 \ge k_2 \ge \cdots \ge k_t$. Let $F_1, F'_1, G_1, G'_1 \in \mathbb{R}_1$ with $F_1 < F'_1, G_1 < G'_1$ and $\max F'_1 = \max G'_1$. Then $\alpha(F_1, F'_1) = \alpha(G_1, G'_1)$ and $\beta(F_1, F'_1) = \beta(G_1, G'_1)$.*

Definition 3.4. *Let $\mathcal{A} \subset \binom{[n]}{k}$ and let $c \in [k]$. We say that $\mathcal{A}$ is* c-sequential *if there exist a set $A \subseteq [n]$ with $|A| = k - c$ and integers $\max A \le a$ and $b \ge a + c$ such that*

$$\mathcal{A} = \{A_r : r \in [b - a - c + 1]\}, \qquad A_r := A \sqcup [a + r, a + r + c - 1].$$

Thus a c-sequential family is obtained by keeping the $(k-c)$-set A fixed and moving a block of c consecutive elements one step to the right. We write $A_r \overset{c}{\prec} A_{r+1}$ for consecutive members. If the final moving block ends at n, we write $A_r \xrightarrow{c} A_q$ for $r < q$.

Lemma 3.5 (Huang-Peng [15])**.** *Let $n \ge k_1 + k_2$ and $k_1 \ge k_2 \ge \cdots \ge k_t$. Let $A_1, B_1, C_1, D_1 \in \mathbb{R}_1$. Suppose that A_1, B_1 are c-sequential and C_1, D_1 are c-sequential with $\max A_1 = \max C_1$ and $\max B_1 = \max D_1$. Then $\alpha(A_1, B_1) = \alpha(C_1, D_1)$ and $\beta(A_1, B_1) = \beta(C_1, D_1)$.*

Proposition 3.6 (Huang-Peng [15])**.** *Suppose that $n \ge k_1 + k_2$, $k_1 \ge k_2 \ge \cdots \ge k_t$ are positive integers. Let $m_i = \min\{k_j : j \in [t] \setminus \{i\}\}$. For each $s \in [m_i]$ and each $i \in [t]$, let*

$$f_i(\{s\}) = \binom{n}{k_i} - \binom{n-s}{k_i} + \sum_{j \ne i} \binom{n-s}{k_j - s}.$$

Then

$$f_1(\{s\}) = \max\{f_j(\{s\}) : j \in [t]\} \text{ for each } s \in [k_t],$$
$$f_t(\{k_{t-1}\}) \le \max\{f_1(\{1\}), f_1(\{k_t\})\}.$$

The next theorem is the global consequence of the local analysis above. In the present paper it will be used only with $d_1 = \cdots = d_t = 1$. We retain the weighted statement because its equality characterization is needed in this paper.

Theorem 3.7 (Huang-Peng [15])**.** *Let $n, t \ge 2$, $k_1, k_2, \ldots, k_t$ be positive integers and $d_1, d_2, \ldots, d_t$ be positive numbers. Let $\mathcal{A}_1 \subseteq \binom{[n]}{k_1}, \mathcal{A}_2 \subseteq \binom{[n]}{k_2}, \ldots, \mathcal{A}_t \subseteq \binom{[n]}{k_t}$ be non-empty pairwise cross-intersecting families with $|\mathcal{A}_i| \ge \binom{n-1}{k_i-1}$ for some $i \in [t]$. Let m_i be the minimum integer among k_j, where $j \in [t] \setminus \{i\}$. If $n \ge k_i + k_j$ for all $j \in [t] \setminus \{i\}$, then*

$$\sum_{j=1}^{t} d_j |\mathcal{A}_j| \le \max\left\{ d_i \binom{n}{k_i} - d_i \binom{n-m_i}{k_i} + \sum_{j \ne i} d_j \binom{n-m_i}{k_j - m_i},\ \sum_{j=1}^{t} d_j \binom{n-1}{k_j - 1} \right\}, \tag{4}$$

the equality holds if and only if one of the following holds.
(i) $d_i\binom{n}{k_i} - d_i\binom{n-m_i}{k_i} + \sum_{j \ne i} d_j \binom{n-m_i}{k_j - m_i} \ge \sum_{j=1}^{t} d_j \binom{n-1}{k_j-1}$, and there is some m_i-element set $T \subseteq [n]$ such that $\mathcal{A}_i = \{F \in \binom{[n]}{k_i} : F \cap T \ne \emptyset\}$ and $\mathcal{A}_j = \{F \in \binom{[n]}{k_j} : T \subseteq F\}$ for each $j \in [t] \setminus \{i\}$.
(ii) $d_i\binom{n}{k_i} - d_i\binom{n-m_i}{k_i} + \sum_{j \ne i} d_j \binom{n-m_i}{k_j - m_i} \le \sum_{j=1}^{t} d_j \binom{n-1}{k_j-1}$, and there is some $a \in [n]$ such that $\mathcal{A}_j = \{F \in \binom{[n]}{k_j} : a \in F\}$ for each $j \in [t]$.
(iii) $n = k_i + k_j$ holds for every $j \in [t] \setminus \{i\}$ and $\sum_{j \ne i} d_j > d_i$. Let $k_j = k$ for all $j \ne i$. If $t = 2$, then $\mathcal{A}_{3-i} \subseteq \binom{[n]}{k_{3-i}}$ with $|\mathcal{A}_{3-i}| = \binom{n-1}{k_{3-i}-1}$ and $\mathcal{A}_i = \binom{[n]}{k_i} \setminus \overline{\mathcal{A}_{3-i}}$. If $t \ge 3$, then for each $j \in [t] \setminus \{i\}$, $\mathcal{A}_j = \mathcal{A}$ and $\mathcal{A}_i = \binom{[n]}{k_i} \setminus \overline{\mathcal{A}}$, where $\mathcal{A}$ is a k-uniform intersecting family with size $|\mathcal{A}| = \binom{n-1}{k-1}$.
(iv) $n = k_i + k_j$ holds for every $j \in [t] \setminus \{i\}$ and $\sum_{j \ne i} d_j = d_i$. Let $k_j = k$ for all $j \ne i$. If $t = 2$, then $\mathcal{A}_{3-i} \subseteq \binom{[n]}{k_{3-i}}$ with $1 \le |\mathcal{A}_{3-i}| \le \binom{n-1}{k_{3-i}-1}$ and $\mathcal{A}_i = \binom{[n]}{k_i} \setminus \overline{\mathcal{A}_{3-i}}$. If $t \ge 3$, then for each $j \in [t] \setminus \{i\}$, $\mathcal{A}_j = \mathcal{A}$ and $\mathcal{A}_i = \binom{[n]}{k_i} \setminus \overline{\mathcal{A}}$, where $\mathcal{A}$ is a k-uniform intersecting family with size $1 \le |\mathcal{A}| \le \binom{n-1}{k-1}$.

4 Proof of Theorem 1.7

Denote

$$\lambda_1 := \sum_{i=1}^{t} \binom{n-1}{k_i-1},$$
$$\lambda_2 := \sum_{i=1}^{2} \left(\binom{n}{k_i} - \binom{n-k_t}{k_i} \right) + \sum_{i=3}^{t} \binom{n-k_t}{k_i-k_t}.$$

Recall Examples 1.5 and 1.6. Clearly, in Example 1.5, $\sum_{i=1}^{t} |\mathcal{G}_i| = \lambda_1$ and Example 1.6, $\sum_{i=1}^{t} |\mathcal{H}_i| = \lambda_2$. Thus, we have the following remark.

Remark 4.1. *Let $k_1 \geq k_2 \geq \cdots \geq k_t$ and $k_1 + k_3 \leq n < k_1 + k_2$. If $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is an extremal $(n, k_1, \ldots, k_t)$-cross-intersecting system, then*

$$\sum_{i=1}^{t} |\mathcal{F}_i| \geq \max\{\lambda_1,\ \lambda_2\}.$$

The following remark derives from Fact 2.6

Remark 4.2. *Let $d \geq 2$ be an integer. We consider d L-initial families $\mathcal{L}([n], A_1, a_1)$, $\ldots$, $\mathcal{L}([n], A_d, a_d)$ with $|A_1| = a_1, \ldots, |A_d| = a_d$. For some fixed $i \in [d]$, let $S \subseteq [d] \setminus \{i\}$ be the set of all $j \in [d] \setminus \{i\}$ satisfying that $n \geq a_i + a_j$. Suppose that $\{1, n - a_i + 2, \ldots, n\} \preceq A_i$. If $\mathcal{L}(A_i, a_i)$ and $\mathcal{L}(A_j, a_j)$ are cross-intersecting for each $j \in S$, then $\mathcal{L}(A_j, a_j)$, $j \in S$ are pairwise cross-intersecting families since for any $j \in S$, every member of $\mathcal{L}(A_j, a_j)$ contains 1.*

Lemma 4.3. *Let $k_1 \geq k_2 \geq \cdots \geq k_t \geq 2$ and $k_1 + k_3 \leq n < k_1 + k_2$. Suppose that $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is an L-initial extremal $(n, k_1, \ldots, k_t)$-cross-intersecting system with IDs $I_1, I_2, \ldots, I_t$ of $\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_t$, respectively. Then $|\mathcal{F}_1| \geq \binom{n-1}{k_1-1}$ and $|\mathcal{F}_2| \geq \binom{n-1}{k_2-1}$, equivalently, $\{1, n - k_1 + 2, \ldots, n\} \prec I_1$ and $\{1, n - k_2 + 2, \ldots, n\} \prec I_2$.*

Proof. If $|\mathcal{F}_i| < \binom{n-1}{k_i-1}$ for all $i \in [t]$, then $\sum_{i=1}^{t} |\mathcal{F}_i| < \lambda_1$, a contradiction to Remark 4.1. So there is $i \in [t]$ such that $|\mathcal{F}_i| \geq \binom{n-1}{k_i-1}$.

First, we consider the case: $i \in [3, t]$ and $I_j \not\preceq \{1, n - k_j + 2, \ldots, n\}$ for each $j \in [2]$. Since $k_1 \geq k_2 \geq \cdots \geq k_t \geq 2$ and $n \geq k_1 + k_3$, $n \geq k_i + k_j$ for all $j \in [t] \setminus \{i\}$. Let $m_i = \min\{k_j : j \in [t] \setminus \{i\}\}$. Taking $d_1 = d_2 = \cdots = d_t = 1$ in Theorem 3.7, we obtain

$$\sum_{j=1}^{t} |\mathcal{F}_j| = \max\left\{ \sum_{j=1}^{t} \binom{n-1}{k_j-1},\ \binom{n}{k_i} - \binom{n-m_i}{k_i} + \sum_{j \in [t]\setminus\{i\}} \binom{n-m_i}{k_j-m_i} \right\}.$$

Note that $n > k_i + m_i$ and for each $j \in [2]$, $I_j \not\preceq \{1, n - k_j + 2, \ldots, n\}$. We only meet Theorem 3.7(i). Thus,

$$\sum_{j=1}^{t} |\mathcal{F}_j| = \binom{n}{k_i} - \binom{n-m_i}{k_i} + \sum_{j \in [t]\setminus\{i\}} \binom{n-m_i}{k_j-m_i}.$$

Setting $s = m_i$ in Proposition 3.6, we have

$$\binom{n}{k_i} - \binom{n-m_i}{k_i} + \sum_{j \notin \{2,i\}} \binom{n-m_i}{k_j-m_i} \leq \max\left\{ \binom{n}{k_1} - \binom{n-k_t}{k_1} + \sum_{j=3}^{t} \binom{n-k_t}{k_j-k_t}, \sum_{j \neq 2} \binom{n-1}{k_j-1} \right\}.$$

Since $m_i \geq k_t \geq 2$, $\binom{n-m_i}{k_2-m_i} < \binom{n-1}{k_2-1}$ and $\binom{n-m_i}{k_2-m_i} < \binom{n}{k_2} - \binom{n-k_t}{k_2}$. Thus, $\sum_{j=1}^{t} |\mathcal{F}_j| < \max\{\lambda_1, \lambda_2\}$, a contradiction to Remark 4.1.

Thus $i \in [2]$. Without loss of generality, we may assume that $i = 1$. Since $n < k_1 + k_2$, then any two families $\mathcal{G} \subseteq \binom{[n]}{k_1}$ and $\mathcal{H} \subseteq \binom{[n]}{k_2}$ are freely cross-intersecting. Note that for each $j \in [3,t]$, $n \geq k_1 + k_j$ and $\mathcal{F}_1$ is cross-intersecting to $\mathcal{F}_j$. Since $\{1, n-k_1+2, \ldots, n\} \prec I_1$, by Remark 4.2, every member of $\mathcal{F}_j$ contains 1. Since $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal, all k_2-subsets containing 1 are contained in $\mathcal{F}_2$, so $|\mathcal{F}_2| \geq \binom{n-1}{k_2-1}$, as required. □

Definition 4.4. *For a set $A \subset [n]$, we define the corresponding k-set $B \subset [n]$ of A as follows: If A has the k-parity, then let B be the k-parity of A; otherwise, let B be the k-set such that $B < A$.*

Let $k < \ell$, for two sets P, R, if $P < R$ with sizes $|P| = |R| = \ell$, by case analysis on whether P, R have k-parity, we have the following fact.

Fact 4.5. *Let $k < \ell$ be positive integers, let $P, R \in \binom{[n]}{\ell}$, and suppose that $\{1, n-\ell+2, \ldots, n\} \prec P < R$. Then the corresponding k-sets of P and R exist; denote them by P' and R', respectively. Moreover, $P' \prec R'$, and $P' = R'$ if and only if R does not have a k-parity.*

Fact 4.6. *Let $A, A' \in \binom{[n]}{a}$, let $1 \leq p \leq q \leq n-a$, and suppose that A and A' have the same q-partner. Then A and A' also have the same p-partner.*

Proof. If $A = A'$, then there is nothing to prove. Thus we may assume that $A \precneqq A'$. For each $r \in \{p, q\}$, let K_r and K'_r be the r-partners of A and A', respectively. Suppose for contradiction that $K_p \neq K'_p$. By Fact 2.11, we have $K'_p \precneqq K_p$. For $r \in \{p, q\}$ and $X \in \{A, A'\}$, define

$$\mathcal{C}_r(X) := \left\{ C \in \binom{[n]}{r} : C \cap B \neq \emptyset \text{ for every } B \in \mathcal{L}(X, a) \right\}.$$

By Proposition 2.2 and Remark 2.4,

$$\mathcal{C}_r(A) = \mathcal{L}(K_r, r) \quad \text{and} \quad \mathcal{C}_r(A') = \mathcal{L}(K'_r, r).$$

Since $K'_p \precneqq K_p$, we have $K_p \in \mathcal{L}(K_p, p) \setminus \mathcal{L}(K'_p, p) = \mathcal{C}_p(A) \setminus \mathcal{C}_p(A')$. Thus, there exists $R \in \mathcal{L}(A', a)$ such that $K_p \cap R = \emptyset$. Since $K_p \in \mathcal{C}_p(A)$, $R \notin \mathcal{L}(A, a)$. Hence $R \in \mathcal{L}(A', a) \setminus \mathcal{L}(A, a)$. Since $|[n] \setminus R| = n - a \geq q$ and $K_p \subseteq [n] \setminus R$, there exists $K \in \binom{[n]\setminus R}{q}$ such that $K_p \subseteq K$. As $K \cap R = \emptyset$, $K \notin \mathcal{C}_q(A')$. On the other hand, $K_p \in \mathcal{C}_p(A)$ and $K_p \subseteq K$, so K intersects every member of $\mathcal{L}(A, a)$. Therefore, $K \in \mathcal{C}_q(A)$. By assumption, A and A' have the same q-partner, so $K_q = K'_q$. Consequently, $\mathcal{C}_q(A) = \mathcal{L}(K_q, q) = \mathcal{L}(K'_q, q) = \mathcal{C}_q(A')$, contradicting $K \in \mathcal{C}_q(A) \setminus \mathcal{C}_q(A')$. Therefore, $K_p = K'_p$, as desired. □

Proposition 4.7. *Let $k_1 \geq k_2 \geq \cdots \geq k_t \geq 2$ and $k_1 + k_3 \leq n < k_1 + k_2$. Suppose that $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is an L-initial extremal $(n, k_1, \ldots, k_t)$-cross-intersecting system with IDs $I_1, I_2, \ldots, I_t$ of $\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_t$, respectively. Then the following properties hold.*

(i) I_2 is the corresponding k_2-set of I_1;

(ii) For each $i \in [3,t]$, I_i is the k_i-partner of I_1.

Proof. By Lemma 4.3, we have $\{1, n-k_1+2, \ldots, n\} \prec I_1$ and $\{1, n-k_2+2, \ldots, n\} \prec I_2$.

First, we consider the case $I_2 \prec I_1$. Take an arbitrary k_2-set $T_2 \subset [n]$ with $\{1, n-k_2+2, \ldots, n\} \prec T_2 \prec I_1$. For each $i \in [3,t]$, let P_i, Q_i be the k_i-partners of I_1, T_2, respectively. By Fact 2.5, for each $i \in [3,t]$, P_i, Q_i are maximal to I_1, T_2, respectively. Combining with $\{1, n-k_1+2, \ldots, n\} \prec I_1$ and $\{1, n-k_2+2, \ldots, n\} \prec T_2$, we see that both $(\mathcal{F}_1, \mathcal{L}(P_3, k_3), \ldots, \mathcal{L}(P_t, k_t))$ and $(\mathcal{L}(T_2, k_2), \mathcal{L}(Q_3, k_3), \ldots, \mathcal{L}(Q_t, k_t))$ are cross-intersecting systems. Since $n \geq k_1 + k_i \geq k_2 + k_i$ and $T_2 \prec I_1$, by Fact 2.11, $P_i \prec Q_i$. This implies that $(\mathcal{L}(T_2, k_2), \mathcal{L}(P_3, k_3), \ldots, \mathcal{L}(P_t, k_t))$ is a cross-intersecting system. Since T_2 is arbitrary and

$\{1, n-k_2+2, \ldots, n\} \prec I_2 \prec I_1$, $(\mathcal{F}_2, \mathcal{L}(P_3, k_3), \ldots, \mathcal{L}(P_t, k_t))$ is also a cross-intersecting system. Thus, $(\mathcal{F}_1, \mathcal{F}_2, \mathcal{L}(P_3, k_3), \ldots, \mathcal{L}(P_t, k_t))$ is a cross-intersecting system. Since P_i is maximal to I_1 for each $i \in [3,t]$, the cross-intersecting property of $\mathcal{F}_1$ and $\mathcal{F}_i$ gives $I_i \prec P_i$. Since $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal, $I_i = P_i$, i.e., I_i is k_i-partner of I_1, we get (ii). Moreover, I_2 must be the largest k_2-set satisfying $I_2 \prec I_1$ in the lex order. Thus, I_2 is the corresponding k_2-set of I_1, we get (i).

Next suppose that $I_1 \prec I_2$. Let $i \in [3,t]$. By the same argument as in the previous case, with the roles of I_1 and I_2 interchanged—that is, taking an arbitrary k_1-set T_1 with $\{1, n-k_1+2, \ldots, n\} \prec T_1 \prec I_2$ in place of T_2, and taking the k_i-partners with respect to I_2 and T_1 instead of I_1 and T_2 (Fact 2.11 applies here since $n \geq k_1 + k_3 \geq \max\{k_1+k_i, k_2+k_i\}$ for each $i \in [3,t]$; in particular, we may take $T_1 = I_1$, as $\{1, n-k_1+2, \ldots, n\} \prec I_1 \prec I_2$ by Lemma 4.3)—we obtain that I_i is the k_i-partner of I_2 for each $i \in [3,t]$. By Fact 2.11, to prove Proposition 4.7, it suffices to prove that I_1 is the k_1-parity of I_2. To see this, we need the following claim.

Claim 4.8. *(I_2, I_3) is maximal.*

Proof. Suppose on the contrary that (I_2, I_3) is not maximal. Let I_2' be the k_2-partner of I_3. Note that I_3 is the k_3-partner of I_2. Since (I_2, I_3) is not maximal, in view of Fact 2.10, $I_2 \preceq I_2'$ and (I_2', I_3) is maximal. So I_2 and I_2' share the same k_3-partner I_3. By Fact 4.6, they also share the same k_i-partner I_i for each $i \in [3,t]$. Thus $(\mathcal{F}_1, \mathcal{L}(I_2', k_2), \mathcal{F}_3, \ldots, \mathcal{F}_t)$ is cross-intersecting. However, $\mathcal{F}_2 \subsetneq \mathcal{L}(I_2', k_2)$, a contradiction to the fact that $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal. This completes the proof of Claim 4.8. □

Let us continue the proof of Proposition 4.7. Since $n \geq k_1 + k_3$ and $k_1 \geq k_2$, combining Proposition 2.13 and Claim 4.8, I_2 has k_1-parity. Since $I_1 \prec I_2$ and $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal, I_1 is the k_1-parity of I_2. This completes the proof of Proposition 4.7. □

For every $i \in [2]$, we denote

$$\mathcal{R}_i = \left\{ R \in \binom{[n]}{k_i} : \{1, n-k_i+2, \ldots, n\} \prec R \prec \{k_t, n-k_i+2, \ldots, n\} \right\}.$$

For every $i \in [3,t]$, we denote

$$\mathcal{R}_i = \left\{ R \in \binom{[n]}{k_i} : [k_t] \cup [n-k_i+k_t+1, n] \prec R \prec \{1, n-k_i+2, \ldots, n\} \right\}.$$

As a consequence of Lemma 4.3 and Proposition 4.7, we have the following remark.

Remark 4.9. *Let $k_1 \geq k_2 \geq \cdots \geq k_t \geq 2$ and $k_1 + k_3 \leq n < k_1 + k_2$. If $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is an L-initial extremal $(n, k_1, \ldots, k_t)$-cross-intersecting system with IDs $I_1, I_2, \ldots, I_t$ of $\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_t$, respectively, then for each $i \in [t]$, we have $I_i \in \mathcal{R}_i$.*

Proposition 4.10. *Let $k_1 \geq k_2 \geq \cdots \geq k_t \geq 2$ and $k_1+k_3 \leq n < k_1+k_2$, and let $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ be an L-initial extremal $(n, k_1, \ldots, k_t)$-cross-intersecting system with IDs $I_1, I_2, \ldots, I_t$ of $\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_t$, respectively. If the set I_1 is either $\{1, n-k_1+2, \ldots, n\}$ or $\{k_t, n-k_1+2, \ldots, n\}$, then $\sum_{i=1}^t |\mathcal{F}_i| = \max\{\lambda_1, \lambda_2\}$.*

Proof. By Proposition 4.7 (i), if $I_1 = \{1, n-k_1+2, \ldots, n\}$, then $I_2 = \{1, n-k_2+2, \ldots, n\}$; if $I_1 = \{k_t, n-k_1+2, \ldots, n\}$, then $I_2 = \{k_t, n-k_2+2, \ldots, n\}$.

Suppose first that $I_1 = \{1, n-k_1+2, \ldots, n\}$. By Proposition 4.7 (ii), for each $i \in [3,t]$, I_i is the k_i-partner of I_1, therefore, $I_i = \{1, n-k_i+2, \ldots, n\}$. Thus $\sum_{i=1}^t |\mathcal{F}_i| = \lambda_1$.

Next, suppose that $I_1 = \{k_t, n-k_1+2, \ldots, n\}$. By Proposition 4.7 (ii) again, for each $i \in [3,t]$, $I_i = \{1, \ldots, k_t, n-k_i+k_t+1, \ldots, n\}$. Thus $\sum_{i=1}^t |\mathcal{F}_i| = \lambda_2$. □

Let $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ be an L-initial extremal $(n,k_1,\dots,k_t)$-cross-intersecting system with $k_1\geq k_2\geq\cdots\geq k_t\geq 2$ and $k_1+k_3\leq n<k_1+k_2$. Remark 4.9 tells us all possible IDs of $\mathcal{F}_i$ for each $i\in[t]$. Proposition 4.10 shows that if the ID of $\mathcal{F}_1$ is the first or the last member of $\mathcal{R}_1$, then $\sum_{i=1}^t|\mathcal{F}_i|$ equals λ_1 or λ_2. A natural question is whether $\sum_{i=1}^t|\mathcal{F}_i|$ is larger than λ_1 and λ_2 if the ID of $\mathcal{F}_1$ is in the intermediate region of $\mathcal{R}_1$. The following theorem shows that the answer is negative, except for an exceptional case.

Theorem 4.11. *Suppose that $t\geq 4$, $k_1\geq k_2\geq\cdots\geq k_t\geq 2$ and $k_1+k_3\leq n<k_1+k_2$. Additionally, suppose that if $t=4$ with $k_1=k_2$ and $k_3=k_4$, then $n>k_1+k_3$. Let $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ be an L-initial $(n,k_1,\dots,k_t)$-cross-intersecting system with ID I_1 of $\mathcal{F}_1$. If $\{1,n-k_1+2,\dots,n\}\precneqq I_1\precneqq\{k_t,n-k_1+2,\dots,n\}$, then $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ is not extremal.*

To prove Theorem 4.11, our idea is to find an $(n,k_1,\dots,k_t)$-cross-intersecting system $(\mathcal{L}(K_1,k_1),\dots,\mathcal{L}(K_t,k_t))$ with $K_i\in\mathcal{R}_i$ for each $i\in[t]$ such that $\sum_{i=1}^t|\mathcal{L}(K_i,k_i)|>\sum_{i=1}^t|\mathcal{F}_i|$. For this purpose, we are going to show 'unimodality' of the forthcoming function $f(R_1)$, where $R_1\in\mathcal{R}_1$. For this purpose, we need to make some preparations first.

Definition 4.12. *Let s,s',t be integers with $s\geq s'\geq 1$, $t\geq s+1$, $k_1=\cdots=k_{s'}>k_{s'+1}\geq\cdots\geq k_t$, and $k_1+k_{s+1}\leq n<k_{s-1}+k_s$. Suppose that $R_1,R_1'\in\mathcal{R}_1$ with $R_1\prec R_1'$. For each $i\in[s+1,t]$, let R_i,R_i' be the k_i-partners of R_1,R_1', respectively; for each $i\in[2,s]$, let R_i,R_i' be the corresponding k_i-sets of R_1,R_1', respectively. For every $i\in[s]$, we denote*

$$\begin{aligned}\alpha_i(R_1,R_1')&=|\mathcal{L}(R_i',k_i)|-|\mathcal{L}(R_i,k_i)|,\\ \gamma(R_1,R_1')&=\sum_{i=1}^{s}\alpha_i(R_1,R_1'),\\ \delta(R_1,R_1')&=\sum_{i=s+1}^{t}\big(|\mathcal{L}(R_i,k_i)|-|\mathcal{L}(R_i',k_i)|\big).\end{aligned}$$

Comparing two Definitions 4.12 and 3.1, we have the following remark.

Remark 4.13. *Let s,s',t be integers with $s\geq s'\geq 1$, $t\geq s+1$, $k_1=\cdots=k_{s'}>k_{s'+1}\geq\cdots\geq k_t$ and $k_1+k_{s+1}\leq n<k_{s-1}+k_s$. Let $R_1,R_1'\in\mathcal{R}_1$ with $R_1\prec R_1'$. Replacing $k_1,k_2,\dots,k_t$ by $k_1,k_{s+1},\dots,k_t$ in Definition 3.1, we can see that $\delta(R_1,R_1')=\beta(R_1,R_1')$, and for each $i\in[s']$, $\alpha_i(R_1,R_1')=\alpha(R_1,R_1')$.*

Definition 4.14. *Let s,s',t be integers with $s\geq s'\geq 1$, $t\geq s+1$, $k_1=\cdots=k_{s'}>k_{s'+1}\geq\cdots\geq k_t$, and $k_1+k_{s+1}\leq n<k_{s-1}+k_s$. Let $R_1\in\mathcal{R}_1$. For each $i\in[2,s]$, let R_i be the corresponding k_i-set of R_1, and for each $i\in[s+1,t]$, let R_i be the k_i-partner of R_1. Denote*

$$f(R_1)=\sum_{i=1}^{t}|\mathcal{L}(R_i,k_i)|.$$

By Definitions 4.12 and 4.14, for any two sets $R_1,R_1'\in\mathcal{R}_1$ with $R_1\prec R_1'$, we have

$$f(R_1')-f(R_1)=\gamma(R_1,R_1')-\delta(R_1,R_1').\tag{5}$$

Remark 4.15. *Let s,s',t be integers with $s\geq s'\geq 1$, $t\geq s+1$, $k_1=\cdots=k_{s'}>k_{s'+1}\geq\cdots\geq k_t$, and $k_1+k_{s+1}\leq n<k_{s-1}+k_s$. Let $R_1\in\mathcal{R}_1$. For each $i\in[2,s]$, let R_i be the corresponding k_i-set of R_1, and for each $i\in[s+1,t]$, let R_i be the k_i-partner of R_1. Then $(\mathcal{L}(R_1,k_1),\dots,\mathcal{L}(R_t,k_t))$ is an $(n,k_1,\dots,k_t)$-cross-intersecting system.*

Proof. Since $n<k_{s-1}+k_s$, $\mathcal{L}(R_1,k_1),\dots,\mathcal{L}(R_s,k_s)$ are pairwise cross-intersecting. Let $i\in[s]$, $j\in[s+1,t]$, and let $R_{i,j}$ be the k_j-partner of R_i. By Fact 2.5, $\mathcal{L}(R_i,k_i)$ and $\mathcal{L}(R_{i,j},k_j)$

are cross-intersecting. Note that R_j is the k_j-partner of R_1, and either $R_i \prec R_1$ or R_i is the K_i-parity of R_1 (if R_1 has the k_i-party). By Fact 2.11, $R_j \prec R_{i,j}$. Thus $\mathcal{L}(R_i, k_i)$ and $\mathcal{L}(R_j, k_j)$ are cross-intersecting. Since $R_1 \in \mathcal{R}_1$, by Remark 4.2, $\mathcal{L}(R_{s+1}, k_{s+1}), \ldots, \mathcal{L}(R_t, k_t)$ are pairwise cross-intersecting. Therefore, $(\mathcal{L}(R_1, k_1), \ldots, \mathcal{L}(R_t, k_t))$ is an $(n, k_1, \ldots, k_t)$-cross-intersecting system. □

Combining Proposition 4.7 and Remark 4.15, and taking $s = 2$ in Definition 4.14, we can see that if $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is an extremal L-initial $(n, k_1, \ldots, k_t)$-cross-intersecting system with ID I_1 of $\mathcal{F}_1$, then

$$\sum_{i=1}^{t} |\mathcal{F}_i| = f(I_1) = \max\{f(R_1) : R_1 \in \mathcal{R}_1\}.$$

As a consequence, to prove Theorem 4.11, it suffices to prove the following theorem.

Theorem 4.16. *Let s, s', t be integers with $s \geq s' \geq 1$, $t \geq s+1$, $k_1 = \cdots = k_{s'} > k_{s'+1} \geq \cdots \geq k_t$, and $k_1 + k_{s+1} \leq n < k_{s-1} + k_s$. Additionally, suppose that if $n = k_1 + k_t$, then $s' \neq t - s$. Let $R_1 \in \mathcal{R}_1$. If $\{1, n-k_1+2, \ldots, n\} \precneqq R_1 \precneqq \{k_t, n-k_1+2, \ldots, n\}$, then there is $K_1 \in \mathcal{R}_1$ such that $f(K_1) > f(R_1)$.*

To prove the above theorem, we need the following lemma, whose proof will be given in the next section. We emphasize that this lemma will be crucial in our follow-up paper [16] for handling the most general case where $n \geq k_1 + k_t$.

Lemma 4.17. *Let s, s', t be integers with $s \geq s' \geq 1$, $t \geq s+1$, $k_1 = \cdots = k_{s'} > k_{s'+1} \geq \cdots \geq k_t$, and $k_1 + k_{s+1} \leq n < k_{s-1} + k_s$. Let $A_1, B_1, C_1 \in \mathcal{R}_1$ with $A_1 \setminus A_1^{\mathrm{t}} = A \sqcup \{a, a+1\}$, $B_1 = A \sqcup \{a\} \sqcup [n - \ell(B_1) + 1, n]$ and $C_1 = A \sqcup \{a+1\} \sqcup [n - \ell(B_1) + 1, n]$ (where $x < a$ for every $x \in A$). Then $\delta(A_1, B_1) = \delta(B_1, C_1)$ and $\gamma(A_1, B_1) \leq \gamma(B_1, C_1)$.*

We are going to prove Theorem 4.16 by applying Proposition 3.2 and Lemma 4.17.

Proof of Theorem 4.16. Suppose on the contrary that

$$f(R_1) = \max\{f(F_1) : F_1 \in \mathcal{R}_1\}. \tag{6}$$

We may assume that R_1 is the first member of $\mathcal{R}_1$ satisfying (6) and $\{1, n-k_1+2, \ldots, n\} \precneqq R_1 \precneqq \{k_t, n-k_1+2, \ldots, n\}$. We have the following two cases.

Case 1. $\ell(R_1) > 0$. In this case, $R_1^t = [n - \ell(R_1) + 1, n]$. Put

$$\ell := \ell(R_1), \quad b := n - \ell + 1, \quad C := R_1 \setminus R_1^{\mathrm{t}}.$$

We first note that $C \neq \emptyset$. Otherwise, $R_1 = [n - k_1 + 1, n]$. Since $n \geq k_1 + k_t$, we have $k_t < n - k_1 + 1$, and hence $\{k_t, n-k_1+2, \ldots, n\} \precneqq R_1$, contrary to $R_1 \prec \{k_t, n-k_1+2, \ldots, n\}$.

Let $a := \max C$ and define

$$P_1 := R_1 \setminus \{b\} \cup \{a+1\}, \qquad Q_1 := R_1 \setminus \{a\} \cup \{a+1\}.$$

By the definition of $\ell(R_1)$, we have $a+1 \notin R_1$ and $a+1 < b$. It follows that both P_1 and Q_1 are k_1-sets. Moreover, $P_1 \precneqq R_1 \precneqq Q_1$.

We next verify that $P_1, Q_1 \in \mathcal{R}_1$. Since $\{1, n-k_1+2, \ldots, n\} \precneqq R_1$, we have $\min R_1 > 1$. Then $\{1, n-k_1+2, \ldots, n\} \precneqq P_1$. Together with $P_1 \precneqq R_1 \precneqq \{k_t, n-k_1+2, \ldots, n\}$, this gives $P_1 \in \mathcal{R}_1$. To deal with Q_1, first observe that $\min R_1 \leq k_t$, since $R_1 \prec \{k_t, n-k_1+2, \ldots, n\}$. If $|C| \geq 2$, removing $\max C$ does not change the minimum, so $\min Q_1 = \min R_1 \leq k_t$. If $|C| = 1$, then $R_1 = \{a\} \cup [n-k_1+2, n]$. Since $R_1 \precneqq \{k_t, n-k_1+2, \ldots, n\}$, we have $a < k_t$, and hence $\min Q_1 = a+1 \leq k_t$. Since $\{k_t, n-k_1+2, \ldots, n\}$ is the last k_1-set whose minimum element is k_t, it follows that $Q_1 \prec \{k_t, n-k_1+2, \ldots, n\}$. Together with $\{1, n-k_1+2, \ldots, n\} \precneqq R_1 \precneqq Q_1$, this proves $Q_1 \in \mathcal{R}_1$.

By the choice of R_1, we have $f(P_1) < f(R_1)$. In view of (5), $f(R_1) = f(P_1) + \gamma(P_1, R_1) - \delta(P_1, R_1)$. Thus,

$$\gamma(P_1, R_1) > \delta(P_1, R_1) \tag{7}$$

Note that P_1, R_1, Q_1 satisfy the conditions of Lemma 4.17, corresponding to A_1, B_1, C_1, respectively. By Lemma 4.17, we have $\delta(P_1, R_1) = \delta(R_1, Q_1)$ and $\gamma(P_1, R_1) \leq \gamma(R_1, Q_1)$. Combining these with (7), and in view of (5), we have

$$f(Q_1) = f(R_1) + \gamma(R_1, Q_1) - \delta(R_1, Q_1) \geq f(R_1) + \gamma(P_1, R_1) - \delta(P_1, R_1) > f(R_1),$$

a contradiction to (6).

Case 2. $\ell(R_1) = 0$. Let $P_1', Q_1' \in \mathcal{R}_1$ be such that $P_1' < R_1 < Q_1'$. Since $\{1, n-k_1+2, \ldots, n\} \precneqq R_1 \precneqq \{k_t, n-k_1+2, \ldots, n\}$, P_1', Q_1' are contained in $\mathcal{R}_1$. Thus $f(P_1') \leq f(R_1)$. In view of (5), $f(R_1) = f(P_1') + \gamma(P_1', R_1) - \delta(P_1', R_1)$. Thus,

$$\gamma(P_1', R_1) \geq \delta(P_1', R_1) \tag{8}$$

For each $i \in [2, s]$, let P_i', R_i, Q_i' be the corresponding k_i-sets of P_1', R_1, Q_1', respectively; for each $i \in [s+1, t]$, let P_i', R_i, Q_i' be the k_i-partners of P_1', R_1, Q_1', respectively.

Since $\ell(R_1) = 0$ and $k_i < k_1$ for $i \in [s'+1, s]$, by Fact 4.5, we have the following claim.

Claim 4.18. *For each $i \in [s'+1, s]$, we have $P_i' = R_i \prec Q_i'$.*

Note that $P_1' < R_1 < Q_1'$. Thus, Claim 4.18 gives

$$\gamma(P_1', R_1) = s' \leq \gamma(R_1, Q_1'). \tag{9}$$

Since $\max R_1 < \max Q_1'$ and $P_1' < R_1 < Q_1'$, combining Proposition 3.2 and Remark 4.13, we obtain one of the following three cases: $\delta(P_1', R_1) > \delta(R_1, Q_1')$; $\delta(P_1', R_1) = \delta(R_1, Q_1') = 0$; $\delta(P_1', R_1) = \delta(R_1, Q_1') = t - s$, where the last case happens if and only if $n = k_1 + k_t$. So $\delta(P_1', R_1) \geq \delta(R_1, Q_1')$. If $\gamma(P_1', R_1) > \delta(P_1', R_1)$ in (8), then by (9), we have

$$f(Q_1') = f(R_1) + \gamma(R_1, Q_1') - \delta(R_1, Q_1') \geq f(R_1) + \gamma(P_1', R_1) - \delta(P_1', R_1) > f(R_1),$$

a contradiction to (6). Thus $\gamma(P_1', R_1) = \delta(P_1', R_1)$ in (8). If $\delta(P_1', R_1) > \delta(R_1, Q_1')$ or $\gamma(P_1', R_1) < \gamma(R_1, Q_1')$ in (9), then

$$f(Q_1') = f(R_1) + \gamma(R_1, Q_1') - \delta(R_1, Q_1') > f(R_1) + \gamma(P_1', R_1) - \delta(P_1', R_1) \geq f(R_1),$$

a contradiction to (6). Thus

$$t - s = \delta(R_1, Q_1') = \delta(P_1', R_1) = \gamma(P_1', R_1) = \gamma(R_1, Q_1') = s',$$

and $n = k_1 + k_t$. This makes a contradiction to the condition of Theorem 4.16.

The proof of Theorem 4.16 is complete. □

Thus, we complete the proof of Theorem 4.11. We are ready to prove Theorem 1.7.

Proof of Theorem 1.7. Let $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ be an extremal $(n, k_1, \ldots, k_t)$-cross-intersecting system with $k_1 \geq k_2 \geq \cdots \geq k_t$ and $k_1 + k_3 \leq n < k_1 + k_2$.

We first consider the case $k_t = 1$. Suppose that $|\mathcal{F}_t| = s$. Then $\mathcal{F}_t = \{\{1\}, \ldots, \{s\}\}$. Since $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is a cross-intersecting system, then for any $i \in [t-1]$ and any $F \in \mathcal{F}_i$, we have $[s] \subseteq F$. Thus,

$$\sum_{i=1}^{t} |\mathcal{F}_i| = \sum_{i=1}^{t-1} \binom{n-s}{k_i - s} + s \leq \lambda_1,$$

as required. Note that the above equality holds if and only if $s = 1$, that is $\mathcal{F}_t = \{x\}$ for some $x \in [n]$. Therefore, $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is isomorphic to $(\mathcal{G}_1, \ldots, \mathcal{G}_t)$ which is defined in Example 1.5, we are done.

Next, consider the case $t = 3$. Recall that $k_3 \geq 2$, since the case $k_t = 1$ has already been treated. By Theorem 1.2, $|\mathcal{F}_1|+|\mathcal{F}_3| \leq \binom{n}{k_1} - \binom{n-k_3}{k_1} + 1$. Clearly, $|\mathcal{F}_2| \leq \binom{n}{k_2} - \binom{n-k_3}{k_2}$ since every k_2-set in $\mathcal{F}_2$ intersects with every k_3-set in $\mathcal{F}_3$. Thus, $\sum_{i=1}^{3} |\mathcal{F}_i| \leq \lambda_2$, as required. The upper bound can be achieved if and only if $|\mathcal{F}_2| = \binom{n}{k_2} - \binom{n-k_3}{k_2}$ and $|\mathcal{F}_1| + |\mathcal{F}_3| = \binom{n}{k_1} - \binom{n-k_3}{k_1} + 1$. Choose any $A \in \mathcal{F}_3$. We have $\mathcal{F}_2 \subseteq \left\{F \in \binom{[n]}{k_2} : F \cap A \neq \emptyset\right\}$. Then $|\mathcal{F}_2| = \binom{n}{k_2} - \binom{n-k_3}{k_2}$ forces $\mathcal{F}_2 = \left\{F \in \binom{[n]}{k_2} : F \cap A \neq \emptyset\right\}$. We claim that $\mathcal{F}_3 = \{A\}$. Otherwise, take $B \in \mathcal{F}_3 \setminus \{A\}$ and choose $x \in A \setminus B$. Since $n \geq k_2 + k_3$, there exists $C \in \binom{[n]\setminus B}{k_2}$ such that $x \in C$. Then $C \cap A \neq \emptyset$, so $C \in \mathcal{F}_2$, whereas $C \cap B = \emptyset$, contradicting the cross-intersection of $\mathcal{F}_2$ and $\mathcal{F}_3$. Thus $\mathcal{F}_3 = \{A\}$. Therefore, $\mathcal{F}_1 = \{F \in \binom{[n]}{k_1} : F \cap A \neq \emptyset\}$. Therefore $(\mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3)$ is isomorphic to the system in Example 1.6. Conversely, that system clearly attains equality.

Then, we consider the case $t = 4$, $k_1 = k_2$, $k_3 = k_4$ and $n = k_1 + k_3$. In this case, $|\mathcal{F}_1| + |\mathcal{F}_4| \leq \binom{n}{k_1}$ and $|\mathcal{F}_2| + |\mathcal{F}_3| \leq \binom{n}{k_1}$. Thus $\sum_{i=1}^{4} |\mathcal{F}_i| \leq 2\binom{n}{k_1} = \lambda_1 = \lambda_2$, as required. Since $(\mathcal{F}_1, \ldots, \mathcal{F}_4)$ is extremal, $\mathcal{F}_1 = \mathcal{F}_2$ and $\mathcal{F}_3 = \mathcal{F}_4$. Note that $n > k_3 + k_4$ and $n = k_1 + k_3$. Then $\mathcal{F}_3$ is an intersecting family, and $\mathcal{F}_1 = \binom{[n]}{k_1} \setminus \overline{\mathcal{F}_3}$, we are done.

Now, it suffices to prove Theorem 1.7 under the following conditions.

$$k_t \geq 2,\ t \geq 4 \text{ and if } t = 4 \text{ with } k_1 = k_2 \text{ and } k_3 = k_4, \text{ then } n > k_1 + k_3. \tag{10}$$

The quantitative part immediately follows from Theorem 2.1, Proposition 4.10 and Theorem 4.11. We are going to show that $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is isomorphic to Examples 1.5 or 1.6 under conditions in (10).

By Lemma 4.3, Proposition 4.10, Theorem 4.11, and Theorem 2.1, we conclude that: either for each $i \in [t]$, we have $|\mathcal{F}_i| = \binom{n-1}{k_i-1}$ or $|\mathcal{F}_1| = \binom{n}{k_1} - \binom{n-k_t}{k_1}$, $|\mathcal{F}_2| = \binom{n}{k_2} - \binom{n-k_t}{k_2}$ and $|\mathcal{F}_i| = \binom{n-k_t}{k_i-k_t}$ holds for each $i \in [3, t]$. Write

$$s_2 := \binom{n-1}{k_2-1}, \qquad h_2 := \binom{n}{k_2} - \binom{n-k_t}{k_2},$$

and

$$S := \sum_{\substack{i=1 \\ i\neq 2}}^{t} \binom{n-1}{k_i-1}, \qquad T := \binom{n}{k_1} - \binom{n-k_t}{k_1} + \sum_{i=3}^{t} \binom{n-k_t}{k_i-k_t}.$$

Thus $\lambda_1 = S + s_2$ and $\lambda_2 = T + h_2$. Since $k_t \geq 2$ under (10),

$$h_2 - s_2 = \binom{n-1}{k_2} - \binom{n-k_t}{k_2} > 0.$$

Suppose first that

$$|\mathcal{F}_i| = \binom{n-1}{k_i-1} \quad \text{for every } i \in [t].$$

This implies $\lambda_1 \geq \lambda_2$. Consequently, $S - T = (\lambda_1 - \lambda_2) + (h_2 - s_2) > 0$. The subsystem $(\mathcal{F}_1, \mathcal{F}_3, \ldots, \mathcal{F}_t)$ has total size S. By Theorem 1.4, the maximum possible total size of such a subsystem is $\max\{S, T\} = S$. Hence this subsystem is extremal.

Notice that $t - 1 \geq 3$ and $k_1 > k_3$; otherwise $k_1 = k_2 = k_3$, and $n \geq 2k_1 = k_1 + k_2$, contrary to $n < k_1 + k_2$. The equality characterization in Theorem 1.4 therefore implies that there exists $x \in [n]$ such that $\mathcal{F}_i = \left\{F \in \binom{[n]}{k_i} : x \in F\right\}$ for every $i \in [t] \setminus \{2\}$. It remains to determine $\mathcal{F}_2$. Since $n \geq k_2 + k_3$, every member of $\mathcal{F}_2$ contains x. Since $|\mathcal{F}_2| = \binom{n-1}{k_2-1}$, we obtain $\mathcal{F}_2 = \left\{F \in \binom{[n]}{k_2} : x \in F\right\}$. Thus the full system is isomorphic to the system in Example 1.5.

Suppose next that

$$|\mathcal{F}_j| = \binom{n}{k_j} - \binom{n-k_t}{k_j} \quad \text{for every } j \in \{1,2\},$$

and

$$|\mathcal{F}_i| = \binom{n-k_t}{k_i-k_t} \quad \text{for every } i \in [3,t].$$

In particular, $|\mathcal{F}_t| = 1$. Write $\mathcal{F}_t = \{T_0\}$, where $|T_0| = k_t$. Since $\mathcal{F}_1$ and $\mathcal{F}_t$ are cross-intersecting, $\mathcal{F}_1 \subseteq \left\{F \in \binom{[n]}{k_1} : F \cap T_0 \neq \emptyset\right\}$. The two sides have the same size and $|\mathcal{F}_1| = \binom{n}{k_1} - \binom{n-k_t}{k_1}$, and hence $\mathcal{F}_1 = \left\{F \in \binom{[n]}{k_1} : F \cap T_0 \neq \emptyset\right\}$. Similarly, $\mathcal{F}_2 = \left\{F \in \binom{[n]}{k_2} : F \cap T_0 \neq \emptyset\right\}$. Furthermore, since $n \geq k_1 + k_i$ for each $i \in [3,t]$, the structure of $\mathcal{F}_1$ forces each $A \in \mathcal{F}_i$ contains T_0. Then by $|\mathcal{F}_i| = \binom{n-k_t}{k_i-k_t}$, we obtain $\mathcal{F}_i = \left\{F \in \binom{[n]}{k_i} : T_0 \subseteq F\right\}$ for every $i \in [3,t]$. Thus the full system is isomorphic to the system in Example 1.6. This completes the proof. □

5 Proof of Lemma 4.17

By Definition 4.12, we have the following remark.

Remark 5.1. *Let s, s', t be integers with $s \geq s' \geq 1$, $t \geq s+1$, $k_1 = \cdots = k_{s'} > k_{s'+1} \geq \cdots \geq k_t$ and $k_1 + k_{s+1} \leq n < k_{s-1} + k_s$. Suppose that $A_1, B_1, C_1 \in \mathcal{R}_1$ with $A_1 \prec B_1 \prec C_1$. Then for any $i \in [s]$, $\alpha_i(A_1, C_1) = \alpha_i(A_1, B_1) + \alpha_i(B_1, C_1)$, and $\gamma(A_1, C_1) = \gamma(A_1, B_1) + \gamma(B_1, C_1)$, $\delta(A_1, C_1) = \delta(A_1, B_1) + \delta(B_1, C_1)$.*

By the definition of parity, we have the following remark.

Remark 5.2. *Let s, s', t be positive integers with $s \geq s' \geq 1$, $t \geq s+1$, $k_1 = \cdots = k_{s'} > k_{s'+1} \geq \cdots \geq k_t$ and $k_1 + k_{s+1} \leq n < k_{s-1} + k_s$. Then for any $R \in \mathcal{R}_1$ and $i \in [2,s]$, R has the k_i-parity if and only if $|R \setminus R^{\mathrm{t}}| \leq k_i$.*

Claim 5.3. *Let s, s', t be integers with $s \geq s' \geq 1$, $t \geq s+1$, $k_1 = \cdots = k_{s'} > k_{s'+1} \geq \cdots \geq k_t$ and $k_1 + k_{s+1} \leq n < k_{s-1} + k_s$. Let $R_1, R_1' \in \mathcal{R}_1$ with $R_1 < R_1'$ and $R = R_1' \setminus {R_1'}^{\mathrm{t}}$. Then for each $i \in [s]$, we have*

$$\alpha_i(R_1, R_1') = \binom{\ell(R_1')}{k_i - |R|}.$$

In particular, $\alpha_i(R_1, R_1') = 0$ if and only if $\ell(R_1') < k_1 - k_i$. Furthermore, if $\ell(R_1') = 0$, then $\gamma(R_1, R_1') = s'$.

Proof. Let $i \in [s]$. Clearly, if $i \in [s']$, then $\alpha_i(R_1, R_1') = 1 = \binom{\ell(R_1')}{\ell(R_1')} = \binom{\ell(R_1')}{k_i - |R|}$. We next consider $i \in [s'+1, s]$. In this case, $k_i < k_1$. Let R_i and R_i' be the corresponding k_i-sets of R_1 and R_1', respectively. Then $R_i \prec R_i'$, moreover $\alpha_i(R_1, R_1') \geq 0$ and $\alpha_i(R_1, R_1') = 0$ if and only if $R_i = R_i'$. By Fact 4.5, $R_i = R_i'$ if and only if R_1' does not have k_i-parity, i.e., $\ell(R_1') < k_1 - k_i$ in view of Remark 5.2. In particular, if $\ell(R_1') = 0$, then $\alpha_i(R_1, R_1') = 0 = \binom{\ell(R_1')}{k_i - |R|}$. Therefore, $\gamma(R_1, R_1') = s'$, as required. We complete the proof of the case that R_1' does not have k_i-parity.

Next, we assume that R_i' is the k_i-parity of R_1'. So $\ell(R_1') \geq k_1 - k_i$. In this case, since $R_1 < R_1'$, $\ell(R_1) = \ell(R_1') - 1$. We first assume that R_1 does not have k_i-parity. Then $\ell(R_1) \leq k_1 - k_i - 1$. Since $\ell(R_1) = \ell(R_1') - 1$, $\ell(R_1') = k_1 - k_i$. Note that $\ell(R_1') = k_1 - |R|$. So $k_i = |R|$ and $R_i' = R$. Note that $R_1 < R_1'$, $\ell(R_1) = \ell(R_1') - 1$ and R_1 does not have k_i-parity. By the definition of corresponding k_i-set, we obtain $R_i < R_i'$. So $\alpha_i(R_1, R_1') = 1 = \binom{\ell(R_1')}{k_i - |R|}$, as required. At last, we assume that R_i is the k_i-parity of R_1. So $\ell(R_1) \geq 1$. Let $k_1 - k_i = k$. In this case we have

$$R_1' = R \cup [n - \ell(R_1') + 1, n];$$

$$R_1 = R \cup \{n - \ell(R_1')\} \cup [n - \ell(R_1') + 2, n];$$
$$R_i = R \cup \{n - \ell(R_1')\} \cup [n - \ell(R_1') + 2 + k, n];$$
$$R_i' = R \cup [n - \ell(R_1') + 1 + k, n] = R \cup [n - k_i + |R| + 1, n].$$

Let A be the k_i-set such that $R_i < A$. Then

$$A = R \cup \{n - \ell(R_1') + 1, \ldots, n - \ell(R_1') + k_i - |R|\}.$$

Denote $\mathcal{F} = \{F \in \binom{[n]}{k_i} : A \prec F \prec R_i'\}$. Then $\alpha_i(R_1, R_1') = |\mathcal{F}| = \binom{n-(n-\ell(R_1'))}{k_i-|R|} = \binom{\ell(R_1')}{k_i-|R|}$, as required. □

Claim 5.4. *Let s, s', t be integers with $s \geq s' \geq 1$, $t \geq s+1$, $k_1 = \cdots = k_{s'} > k_{s'+1} \geq \cdots \geq k_t$ and $k_1 + k_{s+1} \leq n < k_{s-1} + k_s$. Let $F_1, F_1', G_1, G_1' \in \mathcal{R}_1$ be such that $F_1 < F_1', G_1 < G_1'$ and $\max F_1' = \max G_1'$. Then*

(i) $\delta(F_1, F_1') = \delta(G_1, G_1')$;

(ii) if $\ell(F_1') \leq \ell(G_1')$, then $\gamma(F_1, F_1') \leq \gamma(G_1, G_1')$, and if $\ell(F_1') = \ell(G_1')$, then $\gamma(F_1, F_1') = \gamma(G_1, G_1')$.

Proof. Since $F_1 < F_1', G_1 < G_1'$ and $\max F_1' = \max G_1'$, it follows from Proposition 3.3 and Remark 4.13 that

$$\delta(F_1, F_1') = \delta(G_1, G_1'),$$

and for each $i \in [s']$,

$$\alpha_i(F_1, F_1') = \alpha_i(G_1, G_1'). \tag{11}$$

Note that

$$|F_1' \setminus {F_1'}^{\mathrm{t}}| + \ell(F_1') = |G_1' \setminus {G_1'}^{\mathrm{t}}| + \ell(G_1') = k_1. \tag{12}$$

If $\ell(F_1') \leq \ell(G_1')$, then for each $i \in [s'+1, s]$, it follows from Claim 5.3 and (12) that

$$\alpha_i(F_1, F_1') = \binom{\ell(F_1')}{k_i - |F_1' \setminus {F_1'}^{\mathrm{t}}|} \leq \binom{\ell(G_1')}{k_i - |G_1' \setminus {G_1'}^{\mathrm{t}}|} = \alpha_i(G_1, G_1'). \tag{13}$$

Combining (11) and (13), we obtain

$$\gamma(F_1, F_1') = \sum_{i=1}^{s} \alpha_i(F_1, F_1') \leq \sum_{i=1}^{s} \alpha_i(G_1, G_1') = \gamma(G_1, G_1'), \tag{14}$$

as required. Moreover, if $\ell(F_1') = \ell(G_1')$, then equality holds in (13), therefore, equality holds in (14) as well. This completes the proof of Claim 5.4. □

Claim 5.5. *Let s, s', t be integers with $s > s' \geq 1$, $t \geq s+1$, $k_1 = \cdots = k_{s'} > k_{s'+1} \geq \cdots \geq k_t$ and $k_1 + k_{s+1} \leq n < k_{s-1} + k_s$. Let $d \geq 1$ and $A_1, B_1, C_1, D_1 \in \mathcal{R}_1$. Suppose that A_1, B_1 are d-sequential and C_1, D_1 are d-sequential with $\max A_1 = \max C_1$ and $\max B_1 = \max D_1$. Then $\gamma(A_1, B_1) = \gamma(C_1, D_1)$ and $\delta(A_1, B_1) = \delta(C_1, D_1)$. In particular, if $A_1 \xrightarrow{d} B_1$, $C_1 \xrightarrow{d} D_1$ and $\max A_1 = \max C_1$, then $\gamma(A_1, B_1) = \gamma(C_1, D_1)$ and $\delta(A_1, B_1) = \delta(C_1, D_1)$.*

Proof. By the definitions of A_1, B_1, C_1, D_1, from Lemma 3.5 and Remark 4.13, we have $\delta(A_1, B_1) = \delta(C_1, D_1)$ and $\alpha_i(A_1, B_1) = \alpha_i(C_1, D_1)$ holds for each $i \in [s']$. Next, we only need to show that for each $i \in [s'+1, s]$, $\alpha_i(A_1, B_1) = \alpha_i(C_1, D_1)$. Let $i \in [s'+1, s]$. Denote

$$\mathcal{A} = \Big\{R \in \binom{[n]}{k_1} : A_1 \prec R \prec B_1\Big\},$$
$$\mathcal{B} = \Big\{T \in \binom{[n]}{k_1} : C_1 \prec T \prec D_1\Big\}.$$

Since $\alpha_1(A_1,B_1)=\alpha_1(C_1,D_1)$, $|\mathcal{A}|=|\mathcal{B}|=:h$. Let $\mathcal{A}=\{R_1,R_2,\ldots,R_h\}$ and $\mathcal{B}=\{T_1,T_2,\ldots,T_h\}$, where $R_1\prec R_2\prec\cdots\prec R_h$ and $T_1\prec T_2\prec\cdots\prec T_h$. For any $j\in[h]$, we have $\ell(R_j)=\ell(T_j)$ and $|R_j\setminus R_j^{\mathrm{t}}|=|T_j\setminus T_j^{\mathrm{t}}|$. Thus, by Claim 5.3, for any $j\in[h-1]$, $\alpha_i(R_j,R_{j+1})=\alpha_i(T_j,T_{j+1})$. Furthermore, by Remark 5.1, we conclude that

$$\alpha_i(A_1,B_1)=\sum_{j\in[h-1]}\alpha_i(R_j,R_{j+1})=\sum_{j\in[h-1]}\alpha_i(T_j,T_{j+1})=\alpha_i(C_1,D_1),$$

as required. □

Now, we are ready to prove Lemma 4.17.

Proof of Lemma 4.17. From the definitions of A_1,B_1 and C_1, we have $\ell(C_1)\geq\ell(B_1)>\ell(A_1)\geq 0$. Thus $\max B_1=\max C_1=n$. If $\ell(C_1)>\ell(B_1)$, then $A_1<B_1<C_1$, then by Claim 5.4, we have $\delta(A_1,B_1)=\delta(B_1,C_1)$ and $\gamma(A_1,B_1)\leq\gamma(B_1,C_1)$, we are done. Next we may assume that $\ell(C_1)=\ell(B_1)$. Let A_1' and B_1' be the k_1-sets such that $A_1<A_1'$ and $B_1<B_1'$. Then $\max A_1'=\max B_1'$, $\max B_1=\max C_1=n$. Note that $\ell(C_1)=\ell(B_1)$ implies that $A_1<B_1<C_1$ does not happen. So $A_1'\neq B_1$ and $B_1'\neq C_1$, and A_1', B_1 are $(\ell(A_1)+1)$-sequential, B_1', C_1 are $(\ell(A_1)+1)$-sequential. By Claim 5.5, $\gamma(A_1',B_1)=\gamma(B_1',C_1)$ and $\delta(A_1',B_1)=\delta(B_1',C_1)$. Since $A_1<A_1'$ and $B_1<B_1'$ and $\max A_1'=\max B_1'$, by Claim 5.4, $\gamma(A_1,A_1')\leq\gamma(B_1,B_1')$ and $\delta(A_1,A_1')=\delta(B_1,B_1')$. In view of Remark 5.1, we have

$$\begin{aligned}\gamma(A_1,B_1)&=\gamma(A_1,A_1')+\gamma(A_1',B_1)\leq\gamma(B_1,B_1')+\gamma(B_1',C_1)=\gamma(B_1,C_1),\\ \delta(A_1,B_1)&=\delta(A_1,A_1')+\delta(A_1',B_1)=\delta(B_1,B_1')+\delta(B_1',C_1)=\delta(B_1,C_1),\end{aligned}$$

as required. □

6 Acknowledgements

We are grateful to both reviewers for reading the manuscript very carefully, spending a lot of time in checking the details, and giving us valuable comments to help improve the presentation. We would like to thank Andrey Kupavskii for a number of helpful comments and suggestions. This research is supported in part by the National Natural Science Foundation of China (No. 12571363) and the Natural Science Foundation of Hunan Province (Grant No. 2025JJ30003).

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